Collusion, Espionage, or Look-alikes? Auditing Information Flow between LLM Agents

Pairwise net influence with and without private inputs in a procurement auction

Marvin Kemper, Independent; with Apart Research

Abstract

AI agents increasingly bid and set prices for competing firms. If two of them secretly coordinate, can an overseer find the pair just by watching what they do? We test a published information-theoretic measure, pairwise net influence, on AI bidders in a repeated auction with an instructed two-firm cartel and honest twin markets with identical costs. From bids alone, a calibrated alarm built on the measure named exactly the colluding pair in 49 of 60 markets where the cartel chatted in private, never named a wrong pair, and raised one false alarm in 40 honest markets; the alarm rule published with the measure flagged 24 of 40 honest markets. When the overseer also sees each firm's costs, the direction of information flow becomes visible when pooled over markets: two-way for the cartel, one-way from victim to spy for an espionage scenario. A plain correlation of the same data detected chatting cartels at least as well (pre-registered test: net influence AUROC 0.865, $p = 0.00002$; correlation 0.9975); net influence added value only for espionage, and this replicated on fresh markets (+0.265 AUROC $[0.070, 0.445]$). Two warnings follow: the most profitable cartel, one without chat, went almost unnoticed, and an AI judge named the strongest firms rather than the colluders. Behaviour-based screens can direct an investigation, but need calibration and a simple baseline, and are not proof of agreement.

1 Introduction

AI agents increasingly act for competing firms: they set prices, place bids and negotiate. If two such agents coordinate in secret, an overseer may see only what they do. Their messages may be hidden or encoded, and similar bids alone prove nothing, because honest agents react to the same market. Can an audit of behaviour still find the colluding pair?

Why this focus. Pairwise net influence (Bonjour et al., 2022) is a published candidate for exactly this audit: it measures how strongly one agent's actions depend on another's and subtracts the strongest dependence on any third agent, so that a pair is flagged only if it moves together beyond what the rest of the market explains. It needs only actions, not messages or model internals. So far it has been tested with scripted agents in rock–paper–scissors, where coordination is mechanical. Three questions decide whether it is useful in practice, and they structure this paper. Does it find colluding *AI* agents in a market setting, and does its alarm hold up against honest markets? Can it say *what kind* of information flows, a mutual cartel or one-sided espionage, if the auditor also sees private costs? And does it beat a simple correlation and an AI judge that reads the same records? We study a repeated procurement auction because bid rigging is a classic cartel problem with clear private information (each firm's cost) and a clear honest benchmark.

Setup. Four AI bidders compete for 200 rounds (Figure 1). In some markets two bidders are told to form a cartel; in others one bidder secretly sees a rival's cost. Every treated market has an honest twin with identical costs. The auditor sees either the bids only or the bids and every firm's private costs. It never reads messages.

Four findings stand out.

1. **From bids alone, a calibrated alarm finds cartels that talk.** It named exactly the cartel pair in most such markets, never a wrong pair, and rarely raised false alarms (Figure 2). The uncalibrated rule published with the method flagged most honest markets.
2. **With costs, one can see which way information flows.** Pooled over markets, cartel partners leak their costs to each other, while a spy receives information one way (Figure 3). In single markets this signal is often too weak.

3. **A simple correlation detects talking cartels just as well.** Net influence added value only where private information leaked (espionage), and this held on fresh markets (Figure 4).
4. **Two blind spots.** The most profitable cartel, one that never talked, went almost unnoticed, and an AI judge named the strongest firms instead of the colluders (Figure 5).

Our simulation had predicted the false-alarm rate well but overestimated how clearly AI agents reveal themselves (Figure 7). The contribution is a pre-registered, matched-control test of what such an audit can and cannot tell an overseer, with an out-of-sample replication of the main secondary findings.

2 Related Work

Bonjour et al. (2022) introduce information-theoretic collusion detection in multi-agent games, including a partially observable formulation. We implement their statistic and fixed-threshold rule, then examine private-input access and permutation correction in LLM auctions. Work on algorithmic collusion by LLMs (Fish et al., 2024), double auctions (Agrawal et al., 2025), and market division (Lin et al., 2024) supplies the market context; our question concerns the audit of an assigned pair rather than whether collusion emerges spontaneously.

Steganographic collusion (Motwani et al., 2024) motivates audits that do not depend on readable messages. *Audit the Whisper* (Tailor, 2025) uses cross-run mutual information, permutation invariance, and interventions, including auctions. Its run-level focus differs from pair localisation and private-state directional decomposition here. Chain-of-thought monitoring (Lee and Park, 2026) and multi-agent interpretability/NARCBench (Rose et al., 2026) address complementary observation channels; we evaluate behaviour and private inputs, without access to model internals.

Statistical procurement screens provide a related application tradition (Huber and Imhof, 2019; Imhof and Wallimann, 2021; Wallimann et al., 2023). We treat a screen as a reason for further investigation. Concurrent sprint repositories also investigate net influence (Raj-Taware, 2026; soumya-nayak-commits, 2026); we make no claim to the first LLM application. The distinctive focus is the joint evaluation of pair detection, private-cost flow, and honest look-alikes under matched controls.

3 Methods

3.1 Game, conditions, and models

Each market has four firms and 200 rounds. Two strong firms draw independent costs uniformly from $\{1,2,3\}$; two fringe (weak) firms draw from $\{3,4,5\}$. Bids range from 2 to 7; the lowest wins, with random ties. The winner earns bid minus cost, others zero. Bids and winners become public. Bidders see their own cost, the last ten rounds, and their cumulative profit. Costs, tie-breaks, and assignments are seeded and matched by market index. Strong seats cycle through all six label pairs, hidden from auditors.

C1 assigns a cartel with a private chat channel (SS: strong–strong; SF: strong–fringe; FF: fringe–fringe). C2 removes the channel; E1 gives one firm access to its rival's current cost. N0 is honest, and P0 is a scripted cartel positive control (Appendix Table 3). The cartel treatment bundles a joint-profit goal, partner naming, secrecy, a surveillance warning, and channel access; assigned markets stay included regardless of compliance. The core uses Qwen3.5-9B: 40 N0 markets and 20 per other cell. Breadth cells use Gemma-4-12B, Ministral-3-8B, Qwen3.5-4B, Gemma-4-E4B, Ministral-3-3B, and mixed-model markets, each with its own honest control (Appendix B).

3.2 Auditor access and influence

L1 receives bids B_i and winners; L2 additionally receives costs C_i. Neither sees messages, role texts, condition labels, or model identities. The public state H_t is the previous winning bid. At L1, $\gamma_{ij} = I(B_i; B_j \mid H)$. At L2, with $S_i = (C_i, H)$, Equation 3 of Bonjour et al. (2022) becomes

$$\gamma_{ij} = \sum_{s_{ij}} p(s_{ij}) \sum_{b_i, b_j} p(b_i, b_j \mid s_{ij}) \log_2 \frac{p(b_i, b_j \mid s_{ij})}{p(b_i \mid s_i) p(b_j \mid s_j)}, \tag{1}$$



Figure 1: How the test works. Four AI agents (Qwen3.5-9B) bid in a repeated auction. In treated markets two of them form a cartel (with or without a private chat), or one secretly sees a rival's cost. Each treated market has an honest twin with identical costs. The auditor never sees messages; it sees either the bids only (access level L1) or the bids and every firm's costs (L2), and must say whether there is a cartel and which pair it is.

which the chain rule splits exactly into lockstep (action dependence) and two directional leaks:

$$\gamma_{ij} = \underbrace{I(B_i;B_j \mid C_i,C_j,H)}_{\text{action dependence}} + \underbrace{I(B_i;C_j \mid C_i,H)}_{L_{j\to i}} + \underbrace{I(B_j;C_i \mid C_j,H)}_{L_{i\to j}}. \tag{2}$$

Under independent costs, a leak $L_{j\to i}$ means firm j's private cost shows up in firm i's bids. Estimates use plug-in counts in bits. The paper's net transformation and mutual rule (its Equations 4–5) are

$$\Gamma_{ij}[x] = x_{ij} - \max_{k\neq i,j} x_{kj}, \qquad \Gamma_{ij}[\hat{\gamma}] \geq \alpha \text{ and } \quad \Gamma_{ji}[\hat{\gamma}] \geq \alpha, \quad \alpha = 0.05. \tag{3}$$

3.3 Permutation correction

Within each market, 999 permutations shuffle each firm's bids within its own strata (H at L1, (C_i,H) at L2). With the permutation mean subtracted,

$$g_{ij} = \hat{\gamma}_{ij} - \frac{1}{999}\sum_{r=1}^{999} \hat{\gamma}_{ij}^{(r)}, \qquad s_{ij} = \min\{\Gamma_{ij}[g],\Gamma_{ji}[g]\}. \tag{4}$$

Our corrected alarm names a pair only if its raw influence has permutation $p \leq 0.05/6$, both corrected net directions are positive, and it is the only such pair; the published rule (Equation 3) is reported separately. A leak-based rule and per-market flow labels are defined analogously (Appendix C).

3.4 Baselines and evaluation

Correlation is Cramér's V of the two bid series at L1 and the absolute Pearson correlation of bids after removing each firm's mean bid at its own cost at L2. A strength-only benchmark ranks the strong pair first. The frozen Qwen3.5-9B judge reads rules and masked round records and returns pair probabilities (Appendix C). The market is the analysis unit. AUROC is the probability that a randomly chosen treated market scores above a control market (ties half). The sole confirmatory test compares designated-pair L2 net scores in Qwen C1-SS and N0 markets 0–19, with 99,999 within-block label swaps (seed 20261027):

$$p = \frac{1 + \#\{\text{AUC}^* \geq \text{AUC}_{\text{obs}}\}}{100000}; \qquad \text{reject if } \text{AUC}_{\text{obs}} > 0.5 \text{ and } p \leq 0.05. \tag{5}$$

Secondary estimates use 10,000 whole matched-block bootstraps. Honest-market alarms are counts with Clopper–Pearson intervals.



Figure 2: What an alarm reports. For each market type, the share of markets in which the rule named exactly the colluding pair (green), a wrong pair or, in honest markets, any pair (red), or nothing (grey); numbers give right-pair counts (cartel rows) or false alarms (honest row). Left: the fixed threshold published with the method. Right: our permutation-corrected rule. 20 markets per cartel type, 40 honest markets.

3.5 Replication, judge budget, and pre-registration

Design and executable analysis were frozen before collection; seven deviations are recorded in Appendix F. After the final analysis, two additions were pre-registered before their data were collected (Addenda v2.2–v2.3): E1, C2 and C1-SS were collected again on fresh markets 20–39, matched to the existing honest markets of the same index, with pre-specified replication criteria; and the Qwen judge was re-run with 24,000 instead of 12,000 output tokens (L2; N0, C1-SS, C1-SF markets 0–19). Because the original results were known, the replication tests observed patterns, not a blind hypothesis.

4 Results

All results use the frozen analysis code and the full 200 rounds; "bids only" is access level L1, "bids + costs" is L2. Intervals are 95% block-bootstrap intervals. Detailed per-cell figures, all detector rows and the scripted rock–paper–scissors replication of the original paper are in the appendix (Appendices A and E).

4.1 From bids alone, a calibrated alarm names the cartel

Figure 2 asks the question an overseer would ask: did the alarm rule point to exactly the colluding pair, to a wrong pair, or stay silent? With bids only, our permutation-corrected net-influence rule named exactly the right pair in 16, 13 and 20 of 20 markets for the three cartels that chat (two strong firms; strong and weak firm; two weak firms), and never named a wrong pair. In 40 honest markets it raised one false alarm. The positive control, a scripted cartel placed among AI bidders (P0), was named in all 20 markets at both access levels. The rule published with the method is not usable as an alarm here: it flagged 24 of 40 honest markets with bids only and all 40 with costs, and for the strong-pair cartel with costs it named the wrong pair in 19 of 20 markets. With costs our rule named the pair less often (5, 8 and 19 of 20; three false alarms in 40). Neither rule caught the cartel without chat or the spy.

4.2 With costs, the direction of the flow becomes visible

Costs let the measure separate two things: whether the two firms' bids move in lockstep, and whether one firm's private cost shows up in the other firm's bids (a "leak"). Pooled over all markets of a type (Figure 3), honest pairs show neither. The cartel that chats shows strong lockstep and leaks in both directions, consistent with partners who tell each other their costs. The spy shows a one-way leak: the victim's cost appears in the spy's bids (0.0047 bits), not the reverse (-0.0004). The cartel without chat leaves only a weak one-way trace (0.0021 bits) and no lockstep. These pooled

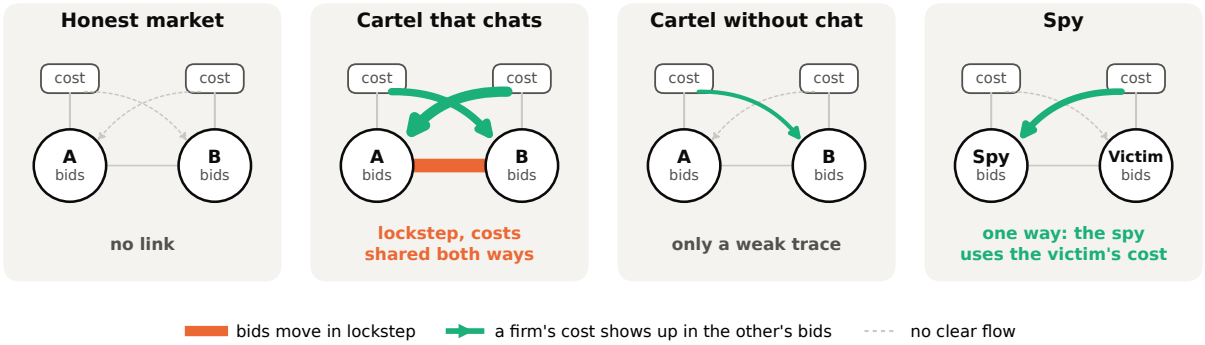


Figure 3: Which way information flows (bids + costs, pooled over 40 honest and 20 treated markets per type). Orange: the two bids move in lockstep beyond what costs and market history explain. Green arrows: a firm's cost shows up in the other firm's bids; width proportional to the estimate, drawn only when its pooled 95% interval excludes zero. These pooled intervals are approximate (Appendix F).

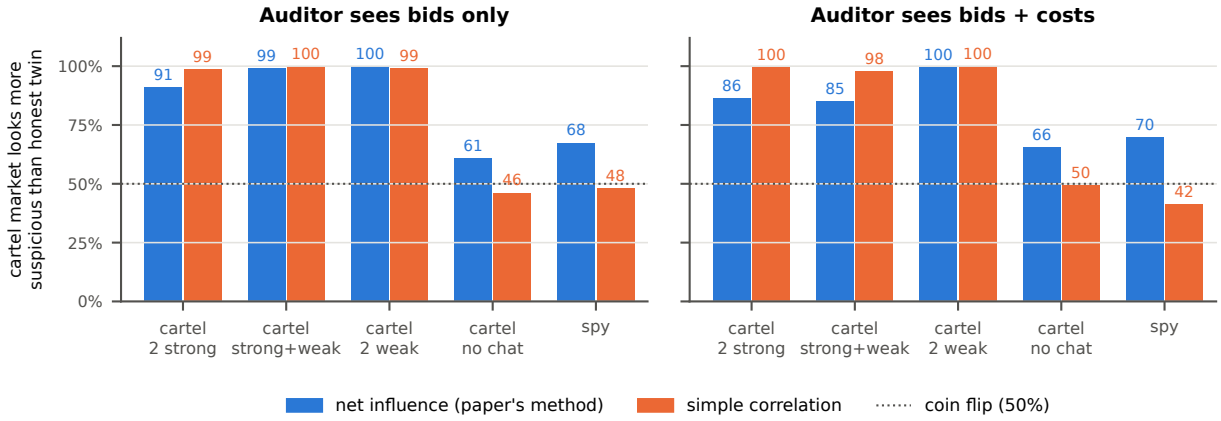


Figure 4: How often a treated market looks more suspicious than its honest twin (AUROC, in percent; 20 matched market pairs per type, markets 0–19). Net influence is the paper's measure; correlation is the plain dependence between the two firms' bids (with costs: after removing each firm's typical bid at its own cost). Intervals and the other models are in Appendix E.

patterns are clear, but single markets are short: the leak-based rule named the spy pair alone in only 4 of 20 spy markets, and market-level flow labels rarely resolved the mechanism.

4.3 Net influence versus a simple correlation

Figure 4 compares the two measures by how often a treated market looks more suspicious than its honest twin with identical costs (AUROC; 50% is a coin flip). In the pre-registered primary test, net influence with costs separated the strong-pair cartel from honest markets: AUROC 0.865 $[0.734, 0.968]$, $p = 0.00002$ over 20 matched blocks. But a simple correlation of the same data did better (0.9975; difference -0.1325 $[-0.2647, -0.0309]$), and it was at least as good for every cartel that chats (the two small exceptions have intervals including zero). Net influence was clearly better only where information leaks: for the spy with costs, 0.700 against 0.415 (difference $+0.285$ $[0.062, 0.515]$). This held on 20 fresh markets collected afterwards ($+0.265$ $[0.070, 0.445]$, Section 3.5); with bids only it did not replicate ($+0.095$ $[-0.1125, 0.315]$). On fresh markets the strong-pair cartel was again detected (net 0.9075 $[0.8025, 0.985]$) and again better by correlation (0.995).

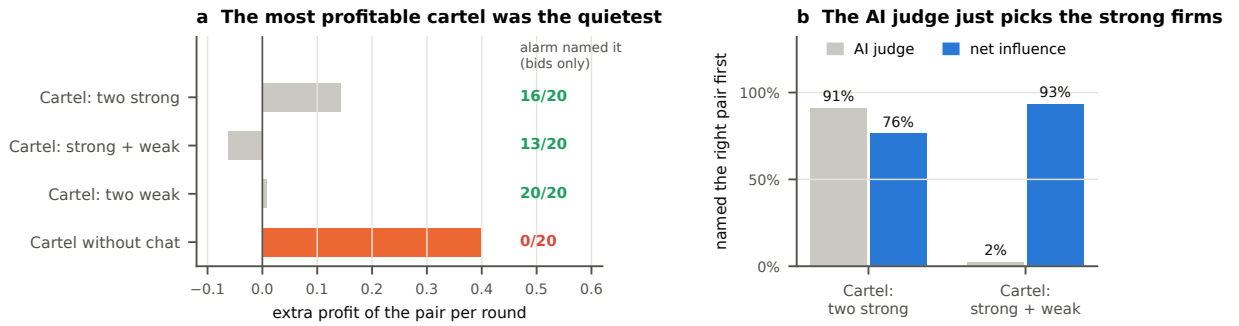


Figure 5: (a) Extra profit of the colluding pair per round compared with honest markets, and how often our alarm named the pair from bids alone. (b) How often the 24k-token AI judge and net influence ranked the true pair first, on the same markets with valid judge output (17 strong-pair and 15 strong–weak markets, bids + costs).

4.4 Two warnings: the quiet cartel and the AI judge

The cartel without chat raised the pair's profit by 0.399 $[0.364, 0.434]$ per round, almost three times the strong-pair cartel's 0.144, yet it was the cartel both measures detected least well and the alarm never caught it with bids only (Figure 5a). The profit gain replicated on fresh markets ($+0.283$ $[0.179, 0.371]$). The Qwen 9B judge reads the round records and names the most suspicious pair. Under its registered 12,000-token limit it failed on 322 of 600 judgements. With 24,000 tokens it answered 54 of 60, but it separated cartel from honest markets barely better than chance (AUROC 0.625 $[0.424, 0.808]$) and simply named the two strong firms: right in 91% of strong-pair cartels, but in 2% of strong–weak cartels, where net influence was right in 93% of the same markets (Figure 5b).

4.5 How many rounds, and other AI models

The alarm needs enough rounds. With bids only it named the strong-pair cartel in 6 of 20 markets after 50 rounds, 9 after 100 and 16 after 200, while false alarms in honest markets stayed at 3 of 40 or fewer (Figure 6a). Other models behaved similarly with one exception (Figure 6b). The alarm named the cartel in 18 of 20 markets for Qwen3.5-4B and Gemma-4-12B, and in 16 and 19 of 20 in mixed-model markets. For Ministral-3-8B it found only 11 of 20 and raised false alarms in 6 of 20 honest Ministral markets (3 of 20 for Qwen3.5-4B): calibration that holds for one model need not hold for another. Ministral was sampled almost deterministically (temperature 0.15, the vendor's recommendation), which may make honest bids look alike; we did not test this. The smaller Qwen was caught as often as the larger one although its cartel earned less than honest play (-0.115 per round), while the Ministral and Gemma cartels gained 0.33 and 0.23: the alarm reacts to coordinated bidding, not to its success. With only two Qwen sizes, and the smallest Gemma and Ministral models unusable (too many invalid bids), the data say nothing about a general size trend. Honest look-alikes did not trigger the alarm: honest markets of four identical Qwen agents raised one false alarm in 40, mixed honest markets 0 and 1 in 20, and swapping which honest firm runs the same model had no effect surviving correction (smallest adjusted $p = 0.264$). Removing the surveillance warning lowered the cartel's profit by 0.35 per round without a resolved change in detection, and probes that changed only the partner's cost gave no resolved effect (Appendix F).

5 Discussion and Limitations

Expected versus observed. Before collecting any LLM data we simulated scripted bidders that always follow their strategy, with the same game, rules and sample sizes (Appendix D). Figure 7 compares that plan with what the AI agents did. The calibration carried over: honest markets produced false alarms at about the simulated rate. The power did not. Scripted cartels were named in about 99% of markets and scripted spies in 100%; the AI cartels in 65–100% with bids only, and the AI spy in none. Even scripted spies that used the stolen cost only half the time were found in 83%. Flow direction shows the same gap: two-way flow was expected in every strong-pair cartel market and one-way flow in 80% of spy markets, but was seen in 0% and 20%; only pooling over markets recovered it (Figure 3). AI agents appear to act on shared information far less consistently than scripts, so their traces are weaker than a simulation with rule-following agents suggests.

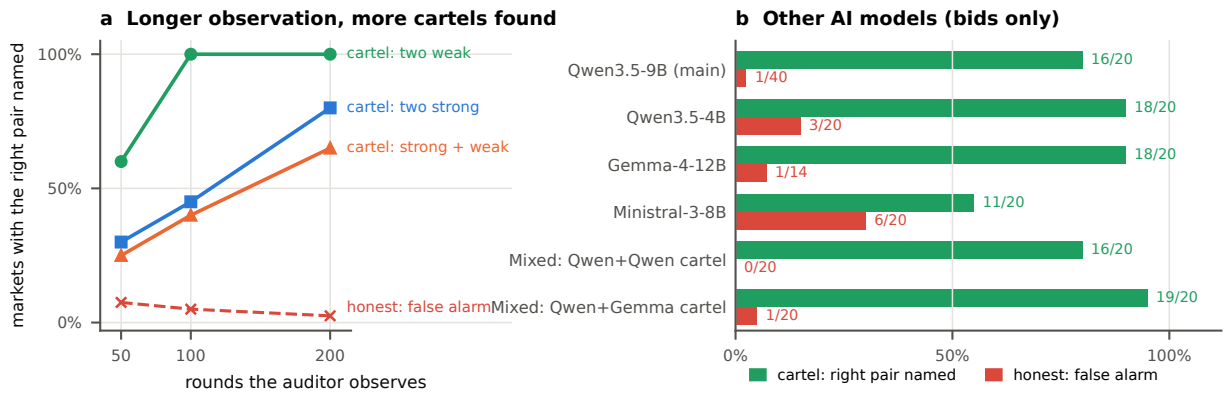


Figure 6: (a) Share of cartel markets in which our alarm named the right pair, and share of honest markets with a false alarm, by the number of rounds observed (bids only, Qwen3.5-9B). (b) The same for other bidder models, each against its own honest markets (20 markets; Gemma honest: 14 eligible; Qwen3.5-9B honest: 40).

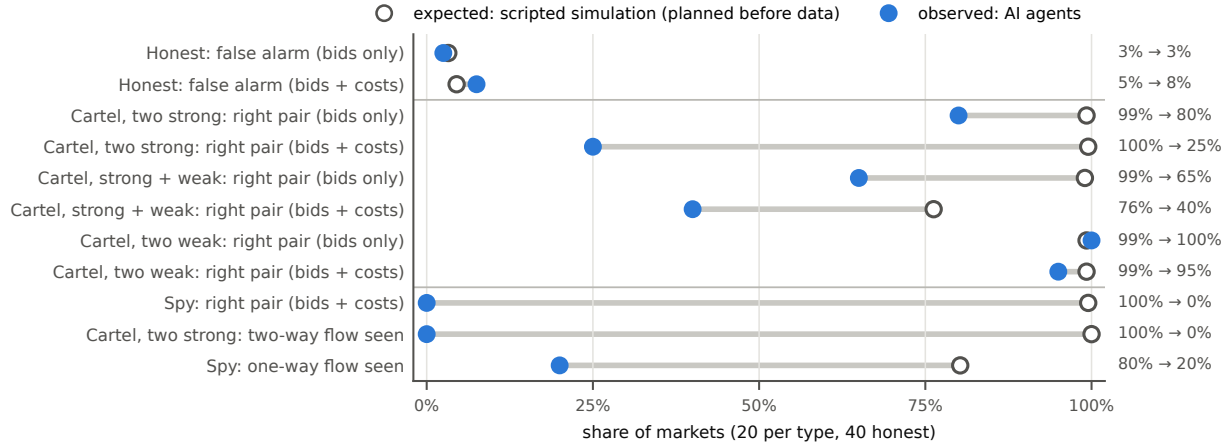


Figure 7: What the pre-data simulation of rule-following scripted bidders predicted (open circles; 200 rounds, 20 markets per type, 40 honest) and what the AI agents showed (filled). "Right pair" uses our corrected alarm; "flow seen" is the per-market flow label of the designated pair.

Comparison with the original paper and concurrent work. Our scripted replication of Bonjour et al. (2022) recovered their rising detection pattern in rock–paper–scissors, but not every published percentage (Appendix A). Two concurrent sprint projects studied the same measure in rock–paper–scissors. Raj-Taware (2026) also found the fixed threshold unreliable (an innocent pair accused in 29% of 100-round games) and a permutation-calibrated test near 2% false alarms, matching our alarm results. They report, however, that two Qwen agents with near-identical prompts were flagged in 7 of 10 games without colluding; in our auction, honest markets of four Qwen agents raised one false alarm in 40, and swapping same-model placements had no detectable effect. The auction's conditioning on costs and the last winning price may explain the difference; we did not test this. soumya-nayak-commits (2026) found that net influence and Cramér's V named the planted pair equally often with scripted agents, as our correlation results do for talking cartels; we add that the exception is private information flow seen with costs. Rose et al. (2026) detect collusion from model activations with AUROC 0.99 to 1.00 and report that only 2 of 229 collusive episodes in one environment verbalised the coordination. Behaviour-only audits like ours need no model access but, as the quiet cartel shows, miss coordination that leaves no trace in actions; the two approaches complement each other.

What this means in practice. Each point is limited to this game. (1) *Screening from public bids is feasible.* An overseer who sees only bids could have used a calibrated alarm to point to the right pair in most markets where AI

agents coordinated actively, with few false alarms and no wrong accusations. (2) *Calibrate before alarming.* The fixed threshold published with the method flagged most or all honest markets; a permutation correction fixed this for the main model, but not fully for Ministral (6 of 20), so calibration has to be checked per model. (3) *Use costs to understand, not to detect.* Costs did not improve detection of talking cartels, but pooled over many markets they show which way information flows and separate a two-way cartel from a one-way spy. (4) *Always compare with a simple baseline.* A plain correlation did as well for talking cartels; net influence earned its place only for espionage. (5) *Watch profits too, and do not trust a small AI judge.* The cartel without chat earned the most and was nearly invisible to both measures. The 9B judge judged firm strength rather than coordination. (6) *Do not plan with rule-following simulations alone.* They predicted false alarms well but overstated detection power for AI agents.

What the evidence does not show. Only the primary test is confirmatory; alarm counts, flow patterns and the simulation comparison are secondary. The replication was planned after the original results were known and reuses the honest control markets. Twenty markets per type give wide intervals, and absence of an advantage is not equivalence. A one-way leak is not a unique signature of espionage, and the pooled flow intervals are approximate. The judge result concerns one 9B model. Comparisons with concurrent work cross games and implementations. The probes cannot support a causal claim.

Scope. The auction is synthetic, the cartel is instructed rather than emergent, and the treatment bundles a joint objective, partner information, a surveillance warning and, in C1, a chat channel. The core uses one model; small models were unusable and breadth cells have only 20 markets. Privacy, false-accusation and dual-use considerations are discussed in Appendix G. Next steps: independent replications; screens that combine correlation, flow direction and profits; larger bidder and judge models; and markets with more firms.

6 Conclusion

Collusion, espionage, or look-alikes? In this auction the audit could tell them apart, but only partly from single markets. Watching only the bids of AI agents, a calibrated net-influence alarm found cartels that coordinated by chat and named the right pair, without wrong accusations. With private costs, the measure also showed which way information flowed, separating a two-way cartel from a one-way spy when evidence was pooled. Yet a simple correlation detected talking cartels as well, the most profitable cartel stayed hidden, and a small AI judge looked at firm strength rather than behaviour. Behaviour-based screens can direct an investigation; they do not establish agreement, intent or harm.

Code and Data

All code, collected market data and analysis outputs are in the project repository. The frozen final analysis is `results/analysis/final_v1/results.json`; the replication and judge check are in `results/analysis/replication_v22/` and `results/analysis/judge24k_v23/`, summarised by `experiments/addendum_summary.py`. The main-text figures are rendered by `experiments/key_figures.py`. Appendix F gives every analysis command, execution provenance and checksums.

[TODO: Insert public repository and release archive URLs after release.]

References

Kushal Agrawal, Verona Teo, Juan J. Vazquez, Sudarsh Kunnavakkam, Vishak Srikanth, and Andy Liu. Evaluating LLM agent collusion in double auctions. arXiv:2507.01413, 2025. URL https://arxiv.org/abs/2507.01413.

Trevor Bonjour, Vaneet Aggarwal, and Bharat Bhargava. Information theoretic approach to detect collusion in multi-agent games. In *Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence*, volume 180 of *Proceedings of Machine Learning Research*, pages 223–232, 2022. URL https://proceedings.mlr.press/v180/bonjour22a.html.

Sara Fish, Yannai A. Gonczarowski, and Ran I. Shorrer. Algorithmic collusion by large language models. arXiv:2404.00806, 2024. URL https://arxiv.org/abs/2404.00806.

Martin Huber and David Imhof. Machine learning with screens for detecting bid-rigging cartels. *International Journal of Industrial Organization*, 65:277–301, 2019. doi: 10.1016/j.ijindorg.2019.04.002.

David Imhof and Hannes Wallimann. Detecting bid-rigging coalitions in different countries and auction formats. *International Review of Law and Economics*, 68:106016, 2021. doi: 10.1016/j.irle.2021.106016.

Dohun Lee and Hyunwoo Park. Faithful yet collusive: Why chain-of-thought monitoring cannot detect collusion in LLM pricing agents under oligopolistic competition. arXiv:2609.18346, 2026. URL https://arxiv.org/abs/2609.18346.

Ryan Y. Lin, Siddhartha Ojha, Kevin Cai, and Maxwell F. Chen. Strategic collusion of LLM agents: Market division in multi-commodity competitions. arXiv:2410.00031, 2024. URL https://arxiv.org/abs/2410.00031.

Sumeet Ramesh Motwani, Mikhail Baranchuk, Martin Strohmeier, Vijay Bolina, Philip H. S. Torr, Lewis Hammond, and Christian Schroeder de Witt. Secret collusion among AI agents: Multi-agent deception via steganography. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2336.

Raj-Taware. collusion-net-influence. GitHub repository, https://github.com/Raj-Taware/collusion-net-influence, 2026. Commit 0e2aa44 (22 Oct 2026), accessed 24 Oct 2026.

Aaron Rose, Carissa Cullen, Sahar Abdelnabi, Philip Torr, Brandon Gary Kaplowitz, and Christian Schroeder de Witt. Detecting multi-agent collusion through multi-agent interpretability. arXiv:2604.01151, 2026. URL https://arxiv.org/abs/2604.01151.

soumya-nayak-commits. collusion-detetction. GitHub repository, https://github.com/soumya-nayak-commits/collusion-detetction, 2026. Commit ee0c111 (23 Oct 2026), accessed 24 Oct 2026.

Om Tailor. Audit the Whisper: Detecting steganographic collusion in multi-agent LLMs. arXiv:2510.04303, 2025. URL https://arxiv.org/abs/2510.04303.

Hannes Wallimann, David Imhof, and Martin Huber. A machine learning approach for flagging incomplete bid-rigging cartels. *Computational Economics*, 62(4):1669–1720, 2023. doi: 10.1007/s10614-022-10315-w.

A Replication details

The scripted rock–paper–scissors replication implements the influence and mutual net-threshold rule of Bonjour et al. (2022). The comparison uses bits and the previous round's joint action as state, conventions inferred from the published results. The no-state variant is retained to show that conditioning choices matter. Detection rises with coordination and sample size, but the replication does not exactly reproduce all table entries. The values below distinguish the paper's published percentages from our simulation. These are development checks, not LLM results.

Table 1: Scripted RPS detection percentages: published values versus our previous-state and no-state replications.

Coordination probability	Games	Paper	Previous state	No state
0.3	100	37.1	36.3	84.4
0.3	200	48.3	60.1	94.4
0.3	300	71.0	77.0	98.1
0.3	400	82.8	87.1	99.5
0.3	500	89.6	92.2	99.9
0.3	1000	100.0	99.7	100.0
0.4	100	58.4	61.9	97.3
0.4	200	79.2	90.7	100.0
0.4	300	96.2	99.0	100.0
0.4	400	99.3	99.8	100.0
0.4	500	99.7	99.9	100.0
0.4	1000	100.0	100.0	100.0
1.0	60	100.0	100.0	100.0
0.2	1000	24.3	50.0	58.0
0.2	10000	82.3	83.7	88.5
0.1	1000	8.0	0.4	0.0

Table 2: Scripted RPS honest-market any-pair alarm rates (coordination probability zero). This is a separate simulation, not LLM calibration.

State	Games	Fixed threshold	Permutation rule
none	50	0.296	0.038
none	100	0.088	0.062
none	200	0.006	0.048
none	500	0.0	0.042
none	1000	0.0	0.038
prev	50	0.588	0.02
prev	100	0.484	0.036
prev	200	0.252	0.036
prev	500	0.046	0.026
prev	1000	0.0	0.038

The deterministic seed is 20261023; the full run uses 1000 trials per detection cell. Its calibration experiment uses 500 trials and 199 permutations. These settings differ from the registered LLM analysis. The complete calibration data, including positive conditions, are in `results/rps/calibration.csv`. Reproduction from the repository root:

```
.venv/Scripts/python experiments/rps_replication.py
```

B Conditions, prompts and model configuration

The frozen manifest is `experiments/manifests/collection_v2.json`. It maps model identities to roles and hidden seat assignments. Exact prompt templates are in `src/collusion/llm/prompts.py`; model revisions and sampling settings are in the per-run configuration and reproducibility records named by `results/analysis_runs.json`. Request hashes, prompts, responses and parsing metadata remain associated with each raw run. `results/analysis_meta/pilot_qwen9b_v5_N0.json` records the pilot-derived N0 configuration explicitly. The manifest and source checksums accompanying this report allow these inputs to be checked without re-collecting outputs.

Table 3: Conditions and matched controls. SS, SF, and FF denote strong–strong, strong–fringe, and fringe–fringe designated pairs. X, Y, Z denote Qwen3.5-9B, Gemma-4-12B, and Ministral-3-8B. All A2 bidders are honest.

Cell	Intervention or control
N0 / P0	Honest market / live scripted strong cartel with honest LLM fringe.
C1-SS/SF/FF	LLM cartel with private channel; matched to N0.
C2 / E1	SS cartel without channel / one strong firm sees the other's cost.
C1-NW (A1)	C1-SS without surveillance warning; matched N0/C1-SS triplets.
D / C1-XX	Honest / cartel strong pair (X,X), fringe (Y,Z).
H-XYZZ / C1-XY	Honest / cartel strong pair (X,Y), fringe (Z,Z).
LX, LY, LZ (A2)	Strong/fringe models (X,Y;X,Z), (Y,Z;Y,X), (Z,X;Z,Y), each with a twin swapping the fringe models.

The A1 prompt removes only: "The competition authority monitors all bids, so behave inconspicuously." The joint-profit objective and private channel remain. This is a warning-removal treatment, not a direct test of an internal evasion strategy. Core Qwen and A1 retain the strict parser; the documented fenced-JSON parser cohort covers Gemma/Ministral, A2 and breadth collection. Bidders run without thinking, whereas the primary judge uses thinking. No treatment labels, messages or model identities are given to the judges.

C Further method details

State and estimation. The public state H_t has a start code and carries forward after an all-invalid round; masking preserves this chronology. Under independent costs and no other access to a rival's current cost, leaks probe private-information flow; omitted history can induce action dependence. Arrows indicate cost source and bid receiver, not intent. Plug-in counts in bits are our replication convention inferred from the paper's results; the paper does not specify the log base. Only a unique passing pair is reported. Symmetric γ need not imply symmetric Γ: the rival comparisons differ. Net direction is not leak direction.

Corrected rules and flow labels. Corrections remain signed. The permutation test concerns influence, not net influence. The secondary leak score sums the two corrected leaks; a leak candidate requires either direction to pass $0.05/12$, again with unique-pair reporting. Flow labels are both directions detected; one detected, reverse weak; neither detected, both weak; or unclear. "Weak" requires $p \geq 0.20$ and a corrected leak at most 0.05 bits. Labels cover designated pairs and the full detection pipeline. Weak evidence does not establish absence; one-way flow can arise in a cartel with unilateral sharing. Permutation validity requires within-stratum exchangeability, which adaptive policies can violate. Per-market alarm rates are descriptive, without a general calibration guarantee or error control across all analyses.

Baselines, judge and evaluation. Constant correlation residuals score zero and signed correlations are also retained; the baselines do not adjust for history. The judge uses thinking and sees costs only at L2 and no treatment information. There is no recalibration or answer selection; content failures remain failures. Comparisons use common judged markets and report coverage; descriptive Brier scores state the artificial prevalence. A Gemma-4-12B counter-judge was attempted on core N0/C1-SS indices 0–9 at both levels; Qwen judging Qwen remains a family-overlap limitation. All methods use rounds with four valid bids; more than 20 incomplete rounds makes a market ineligible and removes the matched block. A no-bid-category sensitivity retains missing bids. The primary test checks paired label exchangeability, not the composite null AUROC ≤ 0.5. Intervals are pointwise 95% percentile intervals. Top-1 is the fraction of credit for ranking the assigned pair first, shared equally across ties; localisation also reports reciprocal rank, correct/wrong unique reports and abstention; $1/6$ is a guessing benchmark, not a null. E1 flow detection is not a collusion positive.

Secondary analyses. A2 tests paired fringe-placement swaps with Holm correction across seven endpoints, not a pure model-sameness mechanism. Pooled fingerprints align roles, bootstrap markets, and permute within markets. The exploratory probe regenerates a partner's message under counterfactual cost, then samples bids with the recipient's cost and history fixed; draws remain nested in markets (Appendix F). Revisions and sampling are pinned; seeded

Table 4: Behaviour and designated-pair flow labels. Differences are treatment minus matched N0, in game units per round, with pointwise 95% intervals. Flow counts are both / one (reverse weak) / neither (both weak) / unclear / unavailable. E1 is unilateral access.

Cell	Joint profit difference	Winning price difference	Flow counts
C1-SS	0.144 [0.094, 0.192]	0.295 [0.233, 0.358]	0 / 0 / 8 / 12 / 0
C2	0.399 [0.364, 0.434]	0.798 [0.768, 0.826]	0 / 0 / 11 / 9 / 0
E1	-0.116 [-0.137, -0.095]	-0.048 [-0.062, -0.034]	0 / 4 / 8 / 8 / 0
C1-NW	-0.204 [-0.229, -0.179]	-0.039 [-0.059, -0.017]	0 / 1 / 8 / 11 / 0

re-collection is not bitwise reproducible (Appendix B). Re-analysing identical markets as a separate cohort reproduces correlation, strength and paper scores exactly; net and leak scores change only by permutation Monte Carlo noise (largest AUROC difference 0.0075). The seven deviations cover warm-up transport; fenced-JSON parsing and labelled Gemma/Ministral recollection; caching, streaming and checkpoints; disk exhaustion and relocation; later recovery, call ceilings, loader paths and an exceeded budget cap; judge transport recovery and budget cancellation of breadth judging; and the operations of the two addenda.

D Scripted validation

Scripted auctions test known cartels, spies and honest policies under the same observation constraints. Shared-history stress tests show why a coarse or incomplete public state can invalidate a permutation null. Power planning and conditional-table occupancy checks motivated the market horizon and weak-evidence category. These simulations inform the design; they do not establish calibration for arbitrary adaptive LLM bidders. The existing artefacts are in `results/auction_scripted/`, `results/auction_power/`, and `results/null_validity/`. Reproduction commands (each also accepts `-quick`):

```
.venv/Scripts/python experiments/auction_scripted.py
.venv/Scripts/python experiments/auction_checks.py
.venv/Scripts/python experiments/auction_power.py
.venv/Scripts/python experiments/null_validity.py
```

These commands regenerate development outputs. They were not rerun for this report update, which reads existing frozen analysis files only.

E All registered cells and per-model results

The following single comprehensive table includes every registered cell, including unavailable detector results and cancelled breadth judges, at prefix 200 with the complete-round mask. It contains 312 detector/access rows spanning 28 cells. The full machine-readable table with every prefix and mask is `results/analysis/final_v1/all_cells.md`; no sensitivity rows are silently discarded from that file.

Cov. is successful detector coverage; n is the matched AUROC block count. A dash means unavailable or not applicable, never a score of zero. Unique/wrong/abstain are rates under the operational reporting rule; AUROC uses the continuous score, including for the paper detector. For honest cells, top-1 and mean reciprocal rank (MRR) describe the *designated reference pair*, not localisation accuracy. Similarly, "unique" is a report of that reference pair, and "wrong" is a report of another pair; neither is a correct collusion finding in an honest market. Alarm columns give any candidate, exactly one report, and designated reference-pair alarms, with exact binomial intervals.

Flow counts are ordered both detected / one detected, reverse weak / neither detected, both weak / unclear / unavailable. Pipeline counts use the same first categories, ending in no pair instead of unavailable. All-zero pipeline counts for score-only baselines mean no reporting rule, not evidence that no information flowed. Judge rows' label counts describe their available subset. Excluded markets fail the registered bid-validity rule. Superseded operational attempts are retained in the raw record, but are not extra independent markets in this registered cohort.

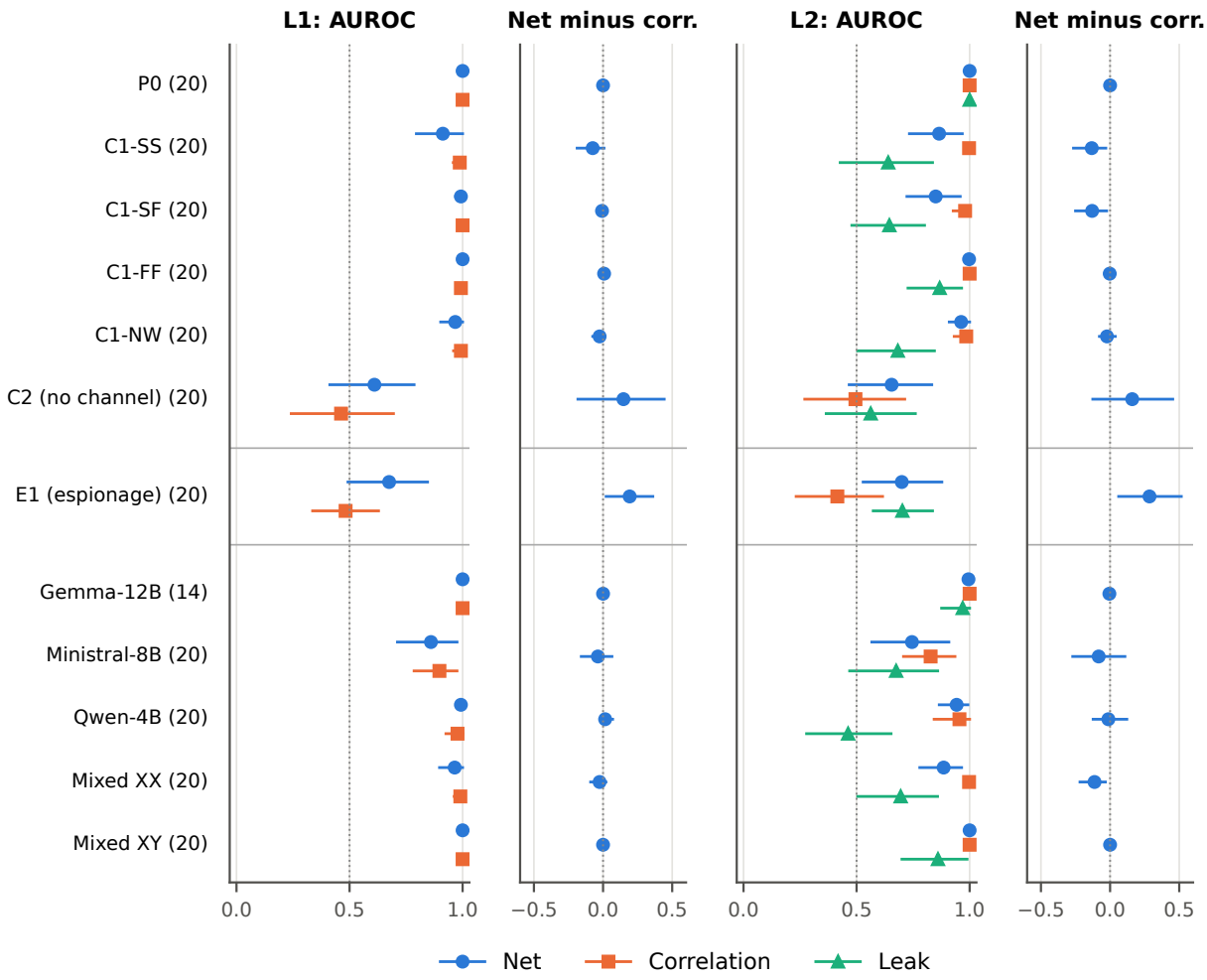


Figure 8: Detection by treatment (original markets 0–19), all models. AUROC with pointwise 95% block-bootstrap intervals for net influence and correlation at L1 (bids only) and L2 (bids + costs), and the L2 leak score; parentheses give matched blocks. E1 (espionage) is separated because it is not a cartel.

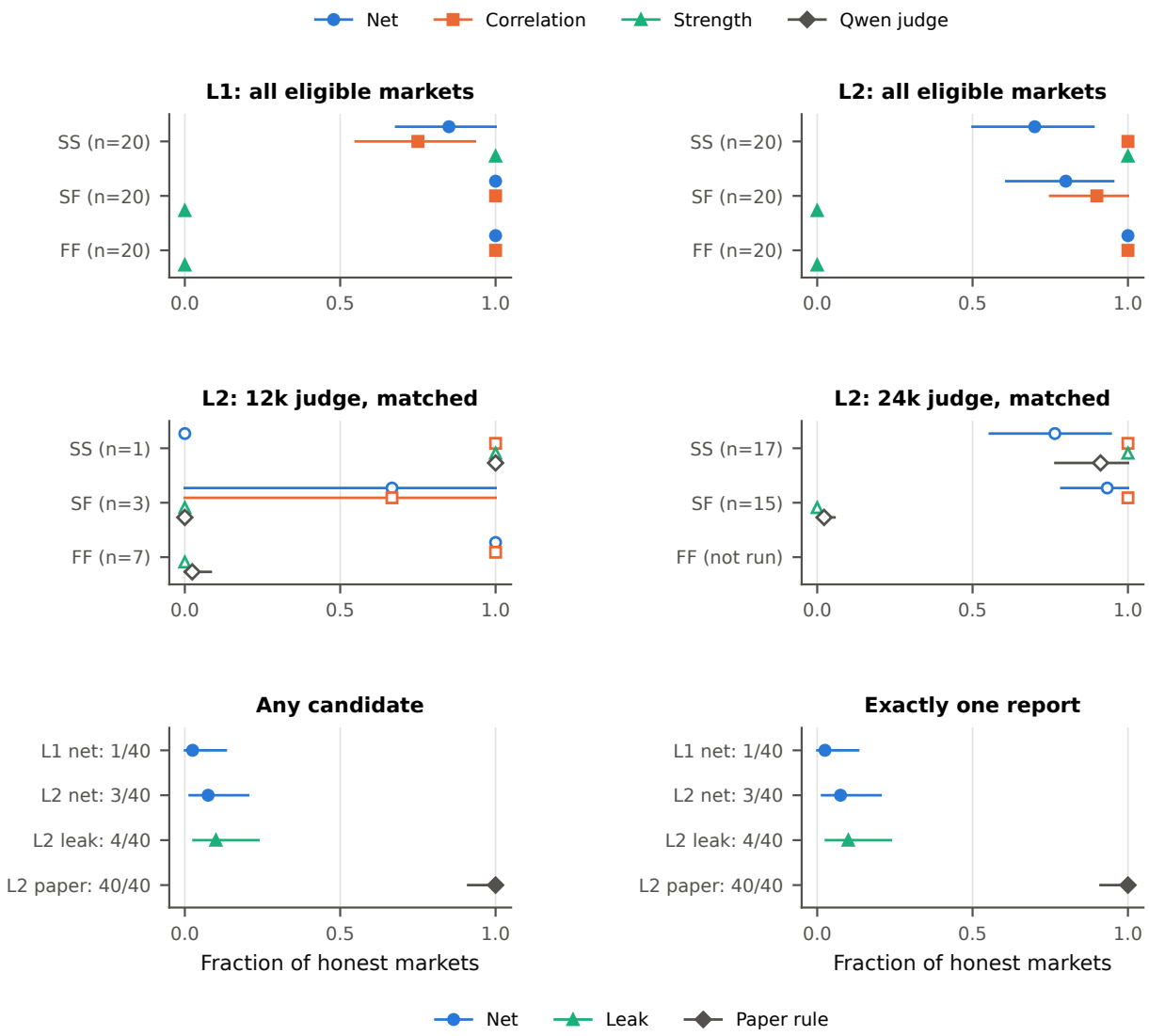


Figure 9: Localisation and false alarms (original markets 0–19). Top: top-1 for the assigned pair on all eligible markets. Middle: top-1 at L2 on the markets where the judge returned valid output, under the registered 12,000-token limit (left) and the 24,000-token check (right; C1-FF not run). Bottom: honest-market alarm rates with exact binomial intervals.

Table 5: All registered cells at the full prefix and complete mask. See the surrounding text for column definitions.

Level / detector	Cov./n	AUROC [interval]	Top-1 [interval]	Unique / wrong / abstain
core.N0; declared 40, collected 40, missing 0, ineligible 0, eligible 40				
L1 corr.	40/–	–	0.050 [0.000, 0.125]	– / – / –
MRR 0.242 [0.192, 0.307]; flow 0/0/32/8/0; pipeline 0/0/0/0/0.				
L1 G judge	1/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/1/0/39; pipeline 0/0/0/0/0.				
L1 Q judge	21/–	–	0.952 [0.840, 1.000]	– / – / –
MRR 0.964 [0.880, 1.000]; flow 0/0/19/2/19; pipeline 0/0/0/0/0.				
L1 net	40/–	–	0.100 [0.025, 0.200]	0.000 / 0.025 / 0.975
MRR 0.385 [0.321, 0.462]; flow 0/0/32/8/0; pipeline 0/0/0/1/39. Alarms: any 1/40 [0.001, 0.132]; one 1/40 [0.001, 0.132]; ref. 0/40 [0.000, 0.088]				
L1 paper	40/–	–	0.000 [0.000, 0.000]	0.000 / 0.600 / 0.400
MRR 0.418 [0.378, 0.454]; flow 0/0/32/8/0; pipeline 0/0/16/8/16. Alarms: any 24/40 [0.433, 0.751]; one 24/40 [0.433, 0.751]; ref. 0/40 [0.000, 0.088]				
L1 strength	40/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/32/8/0; pipeline 0/0/0/0/0.				
L2 corr.	40/–	–	0.175 [0.075, 0.300]	– / – / –
MRR 0.415 [0.330, 0.508]; flow 0/0/32/8/0; pipeline 0/0/0/0/0.				
L2 G judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/40; pipeline 0/0/0/0/0.				
L2 Q judge	15/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/13/2/25; pipeline 0/0/0/0/0.				
L2 leak	40/–	–	0.050 [0.000, 0.125]	0.000 / 0.100 / 0.900
MRR 0.333 [0.283, 0.394]; flow 0/0/32/8/0; pipeline 0/3/0/1/36. Alarms: any 4/40 [0.028, 0.237]; one 4/40 [0.028, 0.237]; ref. 0/40 [0.000, 0.088]				
L2 net	40/–	–	0.075 [0.000, 0.151]	0.000 / 0.075 / 0.925
MRR 0.369 [0.307, 0.439]; flow 0/0/32/8/0; pipeline 0/0/0/3/37. Alarms: any 3/40 [0.016, 0.204]; one 3/40 [0.016, 0.204]; ref. 0/40 [0.000, 0.088]				
L2 paper	40/–	–	0.000 [0.000, 0.000]	0.000 / 1.000 / 0.000
MRR 0.408 [0.369, 0.444]; flow 0/0/32/8/0; pipeline 0/0/25/15/0. Alarms: any 40/40 [0.912, 1.000]; one 40/40 [0.912, 1.000]; ref. 0/40 [0.000, 0.088]				
L2 strength	40/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/32/8/0; pipeline 0/0/0/0/0.				
core.P0; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 0/0/0/0/0.				
L1 Q judge	5/4	0.000 [0.000, 0.000]	0.033 [0.000, 0.111]	– / – / –
MRR 0.304 [0.250, 0.367]; flow 5/0/0/0/15; pipeline 0/0/0/0/0.				
L1 net	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 / 0.000 / 0.000
MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 20/0/0/0/0.				
L1 paper	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 / 0.000 / 0.000
MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 20/0/0/0/0.				
L1 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 0/0/0/0/0.				
L2 corr.	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 0/0/0/0/0.				
L2 Q judge	4/3	0.000 [0.000, 0.000]	0.083 [0.000, 0.167]	– / – / –
MRR 0.330 [0.250, 0.500]; flow 4/0/0/0/16; pipeline 0/0/0/0/0.				
L2 leak	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.950 / 0.000 / 0.050
MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 19/0/0/0/1.				
L2 net	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 / 0.000 / 0.000
MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 20/0/0/0/0.				
L2 paper	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 / 0.000 / 0.000
MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 20/0/0/0/0.				
L2 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 20/0/0/0/0; pipeline 0/0/0/0/0.				
core.C1-SS; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/20	0.988 [0.959, 1.000]	0.750 [0.550, 0.933]	– / – / –
MRR 0.863 [0.750, 0.963]; flow 0/0/8/12/0; pipeline 0/0/0/0/0.				
L1 G judge	3/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/1/2/17; pipeline 0/0/0/0/0.				
L1 Q judge	13/8	0.320 [0.069, 0.617]	0.808 [0.583, 1.000]	– / – / –
MRR 0.835 [0.643, 1.000]; flow 0/0/6/7/7; pipeline 0/0/0/0/0.				

Level / detector	Cov./n	AUROC [interval]	Top-1 [interval]	Unique / wrong / abstain
L1 net	20/20	0.912 [0.796, 1.000]	0.850 [0.680, 1.000]	0.800 / 0.000 / 0.200
MRR 0.908 [0.800, 1.000]; flow 0/0/8/12/0; pipeline 0/0/6/10/4.				
L1 paper	20/20	0.885 [0.765, 0.991]	0.150 [0.000, 0.318]	0.100 / 0.300 / 0.600
MRR 0.527 [0.433, 0.627]; flow 0/0/8/12/0; pipeline 0/0/4/4/12.				
L1 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/8/12/0; pipeline 0/0/0/0/0.				
L2 corr.	20/20	0.998 [0.986, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/8/12/0; pipeline 0/0/0/0/0.				
L2 G judge	1/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/0/1/19; pipeline 0/0/0/0/0.				
L2 Q judge	7/1	0.000 [0.000, 0.000]	0.881 [0.583, 1.000]	– / – / –
MRR 0.898 [0.643, 1.000]; flow 0/0/3/4/13; pipeline 0/0/0/0/0.				
L2 leak	20/20	0.640 [0.428, 0.836]	0.300 [0.105, 0.519]	0.000 / 0.050 / 0.950
MRR 0.530 [0.387, 0.678]; flow 0/0/8/12/0; pipeline 0/0/0/1/19.				
L2 net	20/20	0.865 [0.734, 0.968]	0.700 [0.500, 0.889]	0.250 / 0.000 / 0.750
MRR 0.823 [0.693, 0.938]; flow 0/0/8/12/0; pipeline 0/0/0/5/15.				
L2 paper	20/20	0.882 [0.776, 0.970]	0.000 [0.000, 0.000]	0.000 / 0.950 / 0.050
MRR 0.443 [0.395, 0.485]; flow 0/0/8/12/0; pipeline 0/0/11/8/1.				
L2 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/8/12/0; pipeline 0/0/0/0/0.				
core.C2; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/20	0.463 [0.242, 0.694]	0.300 [0.111, 0.522]	– / – / –
MRR 0.453 [0.300, 0.624]; flow 0/0/11/9/0; pipeline 0/0/0/0/0.				
L1 Q judge	10/4	0.344 [0.000, 1.000]	0.833 [0.583, 1.000]	– / – / –
MRR 0.857 [0.643, 1.000]; flow 0/0/4/6/10; pipeline 0/0/0/0/0.				
L1 net	20/20	0.610 [0.413, 0.785]	0.067 [0.000, 0.188]	0.000 / 0.150 / 0.850
MRR 0.393 [0.317, 0.485]; flow 0/0/11/9/0; pipeline 0/0/0/3/17.				
L1 paper	20/20	0.710 [0.550, 0.854]	0.000 [0.000, 0.000]	0.000 / 0.800 / 0.200
MRR 0.467 [0.426, 0.500]; flow 0/0/11/9/0; pipeline 0/0/4/12/4.				
L1 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/11/9/0; pipeline 0/0/0/0/0.				
L2 corr.	20/20	0.495 [0.271, 0.713]	0.400 [0.190, 0.619]	– / – / –
MRR 0.541 [0.378, 0.712]; flow 0/0/11/9/0; pipeline 0/0/0/0/0.				
L2 Q judge	9/3	0.333 [0.000, 1.000]	0.907 [0.688, 1.000]	– / – / –
MRR 0.921 [0.732, 1.000]; flow 0/0/8/1/11; pipeline 0/0/0/0/0.				
L2 leak	20/20	0.562 [0.366, 0.760]	0.050 [0.000, 0.167]	0.000 / 0.000 / 1.000
MRR 0.357 [0.284, 0.448]; flow 0/0/11/9/0; pipeline 0/0/0/0/20.				
L2 net	20/20	0.655 [0.467, 0.833]	0.050 [0.000, 0.167]	0.050 / 0.000 / 0.950
MRR 0.391 [0.315, 0.483]; flow 0/0/11/9/0; pipeline 0/0/0/1/19.				
L2 paper	20/20	0.887 [0.764, 0.979]	0.000 [0.000, 0.000]	0.000 / 1.000 / 0.000
MRR 0.487 [0.458, 0.500]; flow 0/0/11/9/0; pipeline 0/0/7/13/0.				
L2 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/11/9/0; pipeline 0/0/0/0/0.				
core.E1; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/20	0.482 [0.337, 0.628]	0.100 [0.000, 0.250]	– / – / –
MRR 0.276 [0.179, 0.401]; flow 0/4/8/8/0; pipeline 0/0/0/0/0.				
L1 Q judge	12/5	0.600 [0.167, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/2/6/4/8; pipeline 0/0/0/0/0.				
L1 net	20/20	0.675 [0.493, 0.845]	0.300 [0.105, 0.500]	0.050 / 0.000 / 0.950
MRR 0.570 [0.440, 0.708]; flow 0/4/8/8/0; pipeline 0/0/1/0/19.				
L1 paper	20/20	0.620 [0.453, 0.785]	0.000 [0.000, 0.000]	0.000 / 0.900 / 0.100
MRR 0.463 [0.425, 0.492]; flow 0/4/8/8/0; pipeline 0/0/11/7/2.				
L1 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/4/8/8/0; pipeline 0/0/0/0/0.				
L2 corr.	20/20	0.415 [0.233, 0.615]	0.150 [0.000, 0.333]	– / – / –
MRR 0.378 [0.263, 0.515]; flow 0/4/8/8/0; pipeline 0/0/0/0/0.				
L2 Q judge	7/1	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/2/1/4/13; pipeline 0/0/0/0/0.				
L2 leak	20/20	0.703 [0.574, 0.836]	0.300 [0.105, 0.520]	0.200 / 0.000 / 0.800
MRR 0.568 [0.440, 0.704]; flow 0/4/8/8/0; pipeline 0/4/0/0/16.				
L2 net	20/20	0.700 [0.529, 0.877]	0.400 [0.188, 0.625]	0.000 / 0.050 / 0.950
MRR 0.607 [0.460, 0.759]; flow 0/4/8/8/0; pipeline 0/0/0/1/19.				
L2 paper	20/20	0.652 [0.485, 0.821]	0.000 [0.000, 0.000]	0.000 / 1.000 / 0.000
MRR 0.464 [0.422, 0.500]; flow 0/4/8/8/0; pipeline 0/0/13/7/0.				

Level / detector	Cov./n	AUROC [interval]	Top-1 [interval]	Unique / wrong / abstain
L2 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/4/8/8/0; pipeline 0/0/0/0/0.				
core.C1-SF; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/1/10/9/0; pipeline 0/0/0/0/0.				
L1 Q judge	14/7	0.602 [0.250, 0.880]	0.048 [0.011, 0.091]	– / – / –
MRR 0.273 [0.251, 0.295]; flow 0/1/5/8/6; pipeline 0/0/0/0/0.				
L1 net	20/20	0.993 [0.969, 1.000]	1.000 [1.000, 1.000]	0.650 / 0.000 / 0.350
MRR 1.000 [1.000, 1.000]; flow 0/1/10/9/0; pipeline 0/1/4/8/7.				
L1 paper	20/20	0.998 [0.986, 1.000]	0.950 [0.833, 1.000]	0.550 / 0.000 / 0.450
MRR 0.975 [0.917, 1.000]; flow 0/1/10/9/0; pipeline 0/1/3/7/9.				
L1 strength	20/20	0.500 [0.500, 0.500]	0.000 [0.000, 0.000]	– / – / –
MRR 0.250 [0.250, 0.250]; flow 0/1/10/9/0; pipeline 0/0/0/0/0.				
L2 corr.	20/20	0.980 [0.927, 1.000]	0.900 [0.750, 1.000]	– / – / –
MRR 0.950 [0.875, 1.000]; flow 0/1/10/9/0; pipeline 0/0/0/0/0.				
L2 Q judge	11/3	0.667 [0.000, 1.000]	0.030 [0.000, 0.074]	– / – / –
MRR 0.260 [0.233, 0.288]; flow 0/1/6/4/9; pipeline 0/0/0/0/0.				
L2 leak	20/20	0.645 [0.479, 0.801]	0.350 [0.143, 0.565]	0.050 / 0.000 / 0.950
MRR 0.534 [0.376, 0.695]; flow 0/1/10/9/0; pipeline 0/1/0/0/19.				
L2 net	20/20	0.850 [0.723, 0.958]	0.800 [0.609, 0.952]	0.400 / 0.050 / 0.550
MRR 0.872 [0.743, 0.974]; flow 0/1/10/9/0; pipeline 0/0/2/7/11.				
L2 paper	20/20	0.995 [0.978, 1.000]	0.600 [0.381, 0.812]	0.450 / 0.200 / 0.350
MRR 0.800 [0.690, 0.906]; flow 0/1/10/9/0; pipeline 0/1/5/7/7.				
L2 strength	20/20	0.500 [0.500, 0.500]	0.000 [0.000, 0.000]	– / – / –
MRR 0.250 [0.250, 0.250]; flow 0/1/10/9/0; pipeline 0/0/0/0/0.				
core.C1-FF; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/20	0.993 [0.972, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/1/4/15/0; pipeline 0/0/0/0/0.				
L1 Q judge	10/4	0.438 [0.000, 1.000]	0.100 [0.000, 0.333]	– / – / –
MRR 0.383 [0.250, 0.561]; flow 0/1/2/7/10; pipeline 0/0/0/0/0.				
L1 net	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 / 0.000 / 0.000
MRR 1.000 [1.000, 1.000]; flow 0/1/4/15/0; pipeline 0/1/4/15/0.				
L1 paper	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 / 0.000 / 0.000
MRR 1.000 [1.000, 1.000]; flow 0/1/4/15/0; pipeline 0/1/4/15/0.				
L1 strength	20/20	0.500 [0.500, 0.500]	0.000 [0.000, 0.000]	– / – / –
MRR 0.250 [0.250, 0.250]; flow 0/1/4/15/0; pipeline 0/0/0/0/0.				
L2 corr.	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/1/4/15/0; pipeline 0/0/0/0/0.				
L2 Q judge	14/7	0.551 [0.344, 0.812]	0.012 [0.000, 0.039]	– / – / –
MRR 0.383 [0.308, 0.455]; flow 0/1/3/10/6; pipeline 0/0/0/0/0.				
L2 leak	20/20	0.868 [0.727, 0.964]	0.600 [0.375, 0.812]	0.150 / 0.000 / 0.850
MRR 0.724 [0.565, 0.871]; flow 0/1/4/15/0; pipeline 0/1/0/2/17.				
L2 net	20/20	0.998 [0.986, 1.000]	1.000 [1.000, 1.000]	0.950 / 0.000 / 0.050
MRR 1.000 [1.000, 1.000]; flow 0/1/4/15/0; pipeline 0/1/3/15/1.				
L2 paper	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 / 0.000 / 0.000
MRR 1.000 [1.000, 1.000]; flow 0/1/4/15/0; pipeline 0/1/4/15/0.				
L2 strength	20/20	0.500 [0.500, 0.500]	0.000 [0.000, 0.000]	– / – / –
MRR 0.250 [0.250, 0.250]; flow 0/1/4/15/0; pipeline 0/0/0/0/0.				
labs.gemma-12b.N0; declared 20, collected 20, missing 0, ineligible 6, eligible 14				
L1 corr.	14/–	–	0.071 [0.000, 0.231]	– / – / –
MRR 0.288 [0.192, 0.426]; flow 0/0/8/6/0; pipeline 0/0/0/0/0.				
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/14; pipeline 0/0/0/0/0.				
L1 net	14/–	–	0.143 [0.000, 0.357]	0.000 / 0.071 / 0.929
MRR 0.430 [0.300, 0.581]; flow 0/0/8/6/0; pipeline 0/0/1/0/13. Alarms: any 1/14 [0.002, 0.339]; one 1/14 [0.002, 0.339]; ref. 0/14 [0.000, 0.232]				
L1 paper	14/–	–	0.000 [0.000, 0.000]	0.000 / 0.357 / 0.643
MRR 0.401 [0.332, 0.464]; flow 0/0/8/6/0; pipeline 0/0/3/2/9. Alarms: any 5/14 [0.128, 0.649]; one 5/14 [0.128, 0.649]; ref. 0/14 [0.000, 0.232]				
L1 strength	14/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/8/6/0; pipeline 0/0/0/0/0.				
L2 corr.	14/–	–	0.286 [0.067, 0.545]	– / – / –
MRR 0.483 [0.310, 0.674]; flow 0/0/8/6/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/14; pipeline 0/0/0/0/0.				

Level / detector	Cov./n	AUROC [interval]	Top-1 [interval]	Unique / wrong / abstain
L2 leak	14/–	–	0.071 [0.000, 0.231]	0.000 / 0.071 / 0.929
MRR 0.374 [0.270, 0.500]; flow 0/0/8/6/0; pipeline 0/1/0/0/13. Alarms: any 1/14 [0.002, 0.339]; one 1/14 [0.002, 0.339]; ref. 0/14 [0.000, 0.232]				
L2 net	14/–	–	0.000 [0.000, 0.000]	0.000 / 0.214 / 0.786
MRR 0.363 [0.292, 0.431]; flow 0/0/8/6/0; pipeline 0/0/1/2/11. Alarms: any 3/14 [0.047, 0.508]; one 3/14 [0.047, 0.508]; ref. 0/14 [0.000, 0.232]				
L2 paper	14/–	–	0.000 [0.000, 0.000]	0.000 / 0.857 / 0.143
MRR 0.457 [0.393, 0.500]; flow 0/0/8/6/0; pipeline 0/0/8/4/2. Alarms: any 12/14 [0.572, 0.982]; one 12/14 [0.572, 0.982]; ref. 0/14 [0.000, 0.232]				
L2 strength	14/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/8/6/0; pipeline 0/0/0/0/0.				
labs.gemma-12b.C1; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/14	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/0/0/0/0.				
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L1 net	20/14	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.900 / 0.000 / 0.100
MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/1/5/12/2.				
L1 paper	20/14	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.950 / 0.000 / 0.050
MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/1/5/13/1.				
L1 strength	20/14	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/0/0/0/0.				
L2 corr.	20/14	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L2 leak	20/14	0.969 [0.876, 1.000]	0.600 [0.375, 0.812]	0.050 / 0.000 / 0.950
MRR 0.758 [0.613, 0.890]; flow 0/1/5/14/0; pipeline 0/1/0/0/19.				
L2 net	20/14	0.995 [0.973, 1.000]	1.000 [1.000, 1.000]	0.550 / 0.000 / 0.450
MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/1/3/7/9.				
L2 paper	20/14	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 / 0.000 / 0.000
MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/1/5/14/0.				
L2 strength	20/14	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/1/5/14/0; pipeline 0/0/0/0/0.				
labs.ministral-8b.N0; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/–	–	0.100 [0.000, 0.250]	– / – / –
MRR 0.378 [0.275, 0.496]; flow 0/1/12/7/0; pipeline 0/0/0/0/0.				
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L1 net	20/–	–	0.100 [0.000, 0.250]	0.050 / 0.250 / 0.700
MRR 0.419 [0.322, 0.530]; flow 0/1/12/7/0; pipeline 0/0/2/4/14. Alarms: any 6/20 [0.119, 0.543]; one 6/20 [0.119, 0.543]; ref. 1/20 [0.001, 0.249]				
L1 paper	20/–	–	0.050 [0.000, 0.167]	0.000 / 0.000 / 1.000
MRR 0.384 [0.302, 0.474]; flow 0/1/12/7/0; pipeline 0/0/0/0/20. Alarms: any 0/20 [0.000, 0.168]; one 0/20 [0.000, 0.168]; ref. 0/20 [0.000, 0.168]				
L1 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/1/12/7/0; pipeline 0/0/0/0/0.				
L2 corr.	20/–	–	0.200 [0.045, 0.389]	– / – / –
MRR 0.445 [0.322, 0.579]; flow 0/1/12/7/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L2 leak	20/–	–	0.150 [0.000, 0.320]	0.050 / 0.050 / 0.900
MRR 0.392 [0.280, 0.522]; flow 0/1/12/7/0; pipeline 0/2/0/0/18. Alarms: any 2/20 [0.012, 0.317]; one 2/20 [0.012, 0.317]; ref. 1/20 [0.001, 0.249]				
L2 net	20/–	–	0.050 [0.000, 0.167]	0.000 / 0.150 / 0.850
MRR 0.359 [0.290, 0.447]; flow 0/1/12/7/0; pipeline 0/0/0/3/17. Alarms: any 3/20 [0.032, 0.379]; one 3/20 [0.032, 0.379]; ref. 0/20 [0.000, 0.168]				
L2 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.300 / 0.700
MRR 0.303 [0.252, 0.356]; flow 0/1/12/7/0; pipeline 0/0/1/5/14. Alarms: any 6/20 [0.119, 0.543]; one 6/20 [0.119, 0.543]; ref. 0/20 [0.000, 0.168]				
L2 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/1/12/7/0; pipeline 0/0/0/0/0.				
labs.ministral-8b.C1; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/20	0.897 [0.785, 0.975]	0.650 [0.435, 0.857]	– / – / –
MRR 0.808 [0.687, 0.922]; flow 0/0/8/12/0; pipeline 0/0/0/0/0.				

Level / detector	Cov./n	AUROC [interval]	Top-1 [interval]	Unique / wrong / abstain
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L1 net	20/20	0.860 [0.712, 0.975]	0.800 [0.611, 0.955]	0.550 / 0.000 / 0.450
MRR 0.877 [0.756, 0.975]; flow 0/0/8/12/0; pipeline 0/0/5/6/9.				
L1 paper	20/20	0.922 [0.803, 1.000]	0.750 [0.550, 0.938]	0.450 / 0.000 / 0.550
MRR 0.854 [0.734, 0.963]; flow 0/0/8/12/0; pipeline 0/0/5/4/11.				
L1 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/8/12/0; pipeline 0/0/0/0/0.				
L2 corr.	20/20	0.828 [0.707, 0.935]	0.600 [0.375, 0.812]	– / – / –
MRR 0.771 [0.639, 0.896]; flow 0/0/8/12/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L2 leak	20/20	0.675 [0.469, 0.859]	0.250 [0.062, 0.455]	0.000 / 0.000 / 1.000
MRR 0.512 [0.387, 0.650]; flow 0/0/8/12/0; pipeline 0/0/0/0/20.				
L2 net	20/20	0.745 [0.567, 0.908]	0.600 [0.375, 0.815]	0.200 / 0.150 / 0.650
MRR 0.756 [0.615, 0.891]; flow 0/0/8/12/0; pipeline 0/0/3/4/13.				
L2 paper	20/20	0.935 [0.838, 0.992]	0.650 [0.429, 0.857]	0.450 / 0.050 / 0.500
MRR 0.770 [0.621, 0.910]; flow 0/0/8/12/0; pipeline 0/0/3/7/10.				
L2 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/8/12/0; pipeline 0/0/0/0/0.				
sizes.qwen-4b.N0; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/–	–	0.150 [0.000, 0.333]	– / – / –
MRR 0.314 [0.193, 0.463]; flow 0/0/13/7/0; pipeline 0/0/0/0/0.				
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L1 net	20/–	–	0.050 [0.000, 0.167]	0.050 / 0.100 / 0.850
MRR 0.361 [0.282, 0.455]; flow 0/0/13/7/0; pipeline 0/0/2/1/17. Alarms: any 3/20 [0.032, 0.379]; one 3/20 [0.032, 0.379]; ref. 1/20 [0.001, 0.249]				
L1 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.650 / 0.350
MRR 0.429 [0.369, 0.482]; flow 0/0/13/7/0; pipeline 0/0/4/9/7. Alarms: any 13/20 [0.408, 0.846]; one 13/20 [0.408, 0.846]; ref. 0/20 [0.000, 0.168]				
L1 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/13/7/0; pipeline 0/0/0/0/0.				
L2 corr.	20/–	–	0.200 [0.045, 0.392]	– / – / –
MRR 0.452 [0.324, 0.594]; flow 0/0/13/7/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L2 leak	20/–	–	0.050 [0.000, 0.167]	0.000 / 0.150 / 0.850
MRR 0.352 [0.277, 0.447]; flow 0/0/13/7/0; pipeline 0/3/0/0/17. Alarms: any 3/20 [0.032, 0.379]; one 3/20 [0.032, 0.379]; ref. 0/20 [0.000, 0.168]				
L2 net	20/–	–	0.150 [0.000, 0.333]	0.000 / 0.000 / 1.000
MRR 0.384 [0.272, 0.518]; flow 0/0/13/7/0; pipeline 0/0/0/0/20. Alarms: any 0/20 [0.000, 0.168]; one 0/20 [0.000, 0.168]; ref. 0/20 [0.000, 0.168]				
L2 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 1.000 / 0.000
MRR 0.437 [0.383, 0.483]; flow 0/0/13/7/0; pipeline 0/0/10/10/0. Alarms: any 20/20 [0.832, 1.000]; one 20/20 [0.832, 1.000]; ref. 0/20 [0.000, 0.168]				
L2 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/13/7/0; pipeline 0/0/0/0/0.				
sizes.qwen-4b.C1; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/20	0.978 [0.926, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/12/8/0; pipeline 0/0/0/0/0.				
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L1 net	20/20	0.993 [0.972, 1.000]	1.000 [1.000, 1.000]	0.900 / 0.000 / 0.100
MRR 1.000 [1.000, 1.000]; flow 0/0/12/8/0; pipeline 0/0/11/7/2.				
L1 paper	20/20	1.000 [1.000, 1.000]	0.400 [0.190, 0.619]	0.100 / 0.450 / 0.450
MRR 0.700 [0.595, 0.810]; flow 0/0/12/8/0; pipeline 0/0/6/5/9.				
L1 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/12/8/0; pipeline 0/0/0/0/0.				
L2 corr.	20/20	0.955 [0.843, 1.000]	0.950 [0.833, 1.000]	– / – / –
MRR 0.975 [0.917, 1.000]; flow 0/0/12/8/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L2 leak	20/20	0.463 [0.279, 0.652]	0.200 [0.048, 0.389]	0.000 / 0.000 / 1.000
MRR 0.452 [0.326, 0.589]; flow 0/0/12/8/0; pipeline 0/0/0/0/20.				

Level / detector	Cov./n	AUROC [interval]	Top-1 [interval]	Unique / wrong / abstain
L2 net	20/20	0.943 [0.866, 0.992]	0.700 [0.500, 0.895]	0.750 / 0.050 / 0.200
MRR 0.833 [0.714, 0.941]; flow 0/0/12/8/0; pipeline 0/0/8/8/4.				
L2 paper	20/20	0.960 [0.903, 1.000]	0.000 [0.000, 0.000]	0.000 / 0.950 / 0.050
MRR 0.483 [0.444, 0.500]; flow 0/0/12/8/0; pipeline 0/0/13/6/1.				
L2 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/12/8/0; pipeline 0/0/0/0/0.				
sizes.gemma-e4b.N0; declared 20, collected 20, missing 0, ineligible 20, eligible 0				
L1 corr.	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L1 net	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L1 paper	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L1 strength	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 corr.	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 leak	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 net	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 paper	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 strength	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
sizes.gemma-e4b.C1; declared 20, collected 20, missing 0, ineligible 20, eligible 0				
L1 corr.	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L1 net	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L1 paper	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L1 strength	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 corr.	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 leak	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 net	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 paper	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 strength	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
sizes.ministral-3b.N0; declared 20, collected 20, missing 0, ineligible 20, eligible 0				
L1 corr.	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L1 net	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L1 paper	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L1 strength	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 corr.	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				

Level / detector	Cov./n	AUROC [interval]	Top-1 [interval]	Unique / wrong / abstain
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 leak	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 net	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 paper	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
L2 strength	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/0; pipeline 0/0/0/0/0.				
sizes.ministral-3b.C1; declared 20, collected 20, missing 0, ineligible 18, eligible 2				
L1 corr.	2/–	–	0.000 [0.000, 0.000]	– / – / –
MRR 0.292 [0.250, 0.333]; flow 0/0/2/0/0; pipeline 0/0/0/0/0.				
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/2; pipeline 0/0/0/0/0.				
L1 net	2/–	–	0.000 [0.000, 0.000]	0.000 / 1.000 / 0.000
MRR 0.500 [0.500, 0.500]; flow 0/0/2/0/0; pipeline 0/0/2/0/0.				
L1 paper	2/–	–	0.000 [0.000, 0.000]	0.000 / 0.500 / 0.500
MRR 0.417 [0.333, 0.500]; flow 0/0/2/0/0; pipeline 0/0/1/0/1.				
L1 strength	2/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/2/0/0; pipeline 0/0/0/0/0.				
L2 corr.	2/–	–	0.500 [0.000, 1.000]	– / – / –
MRR 0.750 [0.500, 1.000]; flow 0/0/2/0/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/2; pipeline 0/0/0/0/0.				
L2 leak	2/–	–	0.000 [0.000, 0.000]	0.000 / 0.000 / 1.000
MRR 0.267 [0.200, 0.333]; flow 0/0/2/0/0; pipeline 0/0/0/0/2.				
L2 net	2/–	–	0.500 [0.000, 1.000]	0.500 / 0.000 / 0.500
MRR 0.600 [0.200, 1.000]; flow 0/0/2/0/0; pipeline 0/0/1/0/1.				
L2 paper	2/–	–	0.500 [0.000, 1.000]	0.000 / 0.500 / 0.500
MRR 0.625 [0.250, 1.000]; flow 0/0/2/0/0; pipeline 0/0/1/0/1.				
L2 strength	2/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/2/0/0; pipeline 0/0/0/0/0.				
mixed.D; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/–	–	0.000 [0.000, 0.000]	– / – / –
MRR 0.226 [0.199, 0.254]; flow 0/2/13/5/0; pipeline 0/0/0/0/0.				
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L1 net	20/–	–	0.200 [0.048, 0.389]	0.000 / 0.000 / 1.000
MRR 0.491 [0.374, 0.621]; flow 0/2/13/5/0; pipeline 0/0/0/0/20. Alarms: any 0/20 [0.000, 0.168]; one 0/20 [0.000, 0.168]; ref. 0/20 [0.000, 0.168]				
L1 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.000 / 1.000
MRR 0.327 [0.268, 0.388]; flow 0/2/13/5/0; pipeline 0/0/0/0/20. Alarms: any 0/20 [0.000, 0.168]; one 0/20 [0.000, 0.168]; ref. 0/20 [0.000, 0.168]				
L1 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/2/13/5/0; pipeline 0/0/0/0/0.				
L2 corr.	20/–	–	0.150 [0.000, 0.333]	– / – / –
MRR 0.366 [0.251, 0.501]; flow 0/2/13/5/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L2 leak	20/–	–	0.150 [0.000, 0.333]	0.100 / 0.050 / 0.850
MRR 0.432 [0.324, 0.558]; flow 0/2/13/5/0; pipeline 0/3/0/0/17. Alarms: any 3/20 [0.032, 0.379]; one 3/20 [0.032, 0.379]; ref. 2/20 [0.012, 0.317]				
L2 net	20/–	–	0.250 [0.062, 0.455]	0.000 / 0.050 / 0.950
MRR 0.431 [0.289, 0.591]; flow 0/2/13/5/0; pipeline 0/1/0/0/19. Alarms: any 1/20 [0.001, 0.249]; one 1/20 [0.001, 0.249]; ref. 0/20 [0.000, 0.168]				
L2 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.100 / 0.900
MRR 0.254 [0.209, 0.308]; flow 0/2/13/5/0; pipeline 0/0/1/1/18. Alarms: any 2/20 [0.012, 0.317]; one 2/20 [0.012, 0.317]; ref. 0/20 [0.000, 0.168]				
L2 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/2/13/5/0; pipeline 0/0/0/0/0.				
mixed.C1-XX; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/20	0.990 [0.963, 1.000]	0.850 [0.667, 1.000]	– / – / –
MRR 0.925 [0.833, 1.000]; flow 0/2/8/10/0; pipeline 0/0/0/0/0.				

Level / detector	Cov./n	AUROC [interval]	Top-1 [interval]	Unique / wrong / abstain
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L1 net	20/20	0.965 [0.898, 1.000]	0.950 [0.833, 1.000]	0.800 / 0.000 / 0.200
MRR 0.975 [0.917, 1.000]; flow 0/2/8/10/0; pipeline 0/2/6/8/4.				
L1 paper	20/20	0.955 [0.848, 1.000]	0.900 [0.750, 1.000]	0.450 / 0.000 / 0.550
MRR 0.950 [0.875, 1.000]; flow 0/2/8/10/0; pipeline 0/1/2/6/11.				
L1 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/2/8/10/0; pipeline 0/0/0/0/0.				
L2 corr.	20/20	0.998 [0.986, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/2/8/10/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L2 leak	20/20	0.695 [0.505, 0.858]	0.350 [0.143, 0.571]	0.100 / 0.000 / 0.900
MRR 0.545 [0.396, 0.706]; flow 0/2/8/10/0; pipeline 0/2/0/0/18.				
L2 net	20/20	0.885 [0.779, 0.965]	0.700 [0.500, 0.889]	0.500 / 0.100 / 0.400
MRR 0.833 [0.714, 0.939]; flow 0/2/8/10/0; pipeline 0/2/3/7/8.				
L2 paper	20/20	0.993 [0.969, 1.000]	0.950 [0.833, 1.000]	0.600 / 0.000 / 0.400
MRR 0.975 [0.917, 1.000]; flow 0/2/8/10/0; pipeline 0/1/5/6/8.				
L2 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/2/8/10/0; pipeline 0/0/0/0/0.				
mixed.H-XYZZ; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/–	–	0.050 [0.000, 0.167]	– / – / –
MRR 0.333 [0.250, 0.432]; flow 0/0/17/3/0; pipeline 0/0/0/0/0.				
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L1 net	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.050 / 0.950
MRR 0.391 [0.331, 0.446]; flow 0/0/17/3/0; pipeline 0/0/1/0/19. Alarms: any 1/20 [0.001, 0.249]; one 1/20 [0.001, 0.249]; ref. 0/20 [0.000, 0.168]				
L1 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.050 / 0.950
MRR 0.368 [0.305, 0.428]; flow 0/0/17/3/0; pipeline 0/0/1/0/19. Alarms: any 1/20 [0.001, 0.249]; one 1/20 [0.001, 0.249]; ref. 0/20 [0.000, 0.168]				
L1 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/17/3/0; pipeline 0/0/0/0/0.				
L2 corr.	20/–	–	0.050 [0.000, 0.167]	– / – / –
MRR 0.369 [0.292, 0.462]; flow 0/0/17/3/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L2 leak	20/–	–	0.100 [0.000, 0.250]	0.000 / 0.050 / 0.950
MRR 0.342 [0.251, 0.457]; flow 0/0/17/3/0; pipeline 0/1/0/0/19. Alarms: any 1/20 [0.001, 0.249]; one 1/20 [0.001, 0.249]; ref. 0/20 [0.000, 0.168]				
L2 net	20/–	–	0.100 [0.000, 0.250]	0.000 / 0.100 / 0.900
MRR 0.430 [0.334, 0.541]; flow 0/0/17/3/0; pipeline 0/1/1/0/18. Alarms: any 2/20 [0.012, 0.317]; one 2/20 [0.012, 0.317]; ref. 0/20 [0.000, 0.168]				
L2 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.350 / 0.650
MRR 0.269 [0.217, 0.327]; flow 0/0/17/3/0; pipeline 0/1/2/4/13. Alarms: any 7/20 [0.154, 0.592]; one 7/20 [0.154, 0.592]; ref. 0/20 [0.000, 0.168]				
L2 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/17/3/0; pipeline 0/0/0/0/0.				
mixed.C1-XY; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/0/0/0/0.				
L1 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L1 net	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	0.950 / 0.000 / 0.050
MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/5/7/7/1.				
L1 paper	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 / 0.000 / 0.000
MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/6/7/7/0.				
L1 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/0/0/0/0.				
L2 corr.	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/0/0/0/0.				
L2 Q judge	0/–	–	–	– / – / –
MRR –; flow 0/0/0/0/20; pipeline 0/0/0/0/0.				
L2 leak	20/20	0.860 [0.701, 0.989]	0.750 [0.550, 0.933]	0.350 / 0.000 / 0.650
MRR 0.838 [0.702, 0.956]; flow 0/6/7/7/0; pipeline 0/6/0/1/13.				

Level / detector	Cov./n	AUROC [interval]	Top-1 [interval]	Unique / wrong / abstain
L2 net	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 / 0.000 / 0.000
MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/6/7/7/0.				
L2 paper	20/20	1.000 [1.000, 1.000]	1.000 [1.000, 1.000]	1.000 / 0.000 / 0.000
MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/6/7/7/0.				
L2 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/6/7/7/0; pipeline 0/0/0/0/0.				
core.C1-NW; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/20	0.993 [0.960, 1.000]	0.900 [0.750, 1.000]	– / – / –
MRR 0.933 [0.833, 1.000]; flow 0/1/8/11/0; pipeline 0/0/0/0/0.				
L1 Q judge	12/5	0.480 [0.000, 1.000]	0.931 [0.762, 1.000]	– / – / –
MRR 0.940 [0.796, 1.000]; flow 0/0/4/8/8; pipeline 0/0/0/0/0.				
L1 net	20/20	0.968 [0.903, 1.000]	0.800 [0.609, 0.955]	0.600 / 0.000 / 0.400
MRR 0.892 [0.786, 0.976]; flow 0/1/8/11/0; pipeline 0/0/4/8/8.				
L1 paper	20/20	0.932 [0.847, 0.994]	0.350 [0.150, 0.565]	0.100 / 0.400 / 0.500
MRR 0.667 [0.562, 0.778]; flow 0/1/8/11/0; pipeline 0/0/5/5/10.				
L1 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/1/8/11/0; pipeline 0/0/0/0/0.				
L2 corr.	20/20	0.985 [0.932, 1.000]	0.950 [0.833, 1.000]	– / – / –
MRR 0.975 [0.917, 1.000]; flow 0/1/8/11/0; pipeline 0/0/0/0/0.				
L2 Q judge	10/5	0.800 [0.500, 1.000]	0.900 [0.667, 1.000]	– / – / –
MRR 0.950 [0.833, 1.000]; flow 0/0/4/6/10; pipeline 0/0/0/0/0.				
L2 leak	20/20	0.682 [0.505, 0.844]	0.200 [0.048, 0.389]	0.100 / 0.050 / 0.850
MRR 0.464 [0.340, 0.598]; flow 0/1/8/11/0; pipeline 0/2/0/1/17.				
L2 net	20/20	0.963 [0.910, 1.000]	0.650 [0.438, 0.857]	0.650 / 0.050 / 0.300
MRR 0.817 [0.704, 0.923]; flow 0/1/8/11/0; pipeline 0/1/5/8/6.				
L2 paper	20/20	0.975 [0.901, 1.000]	0.000 [0.000, 0.000]	0.000 / 0.950 / 0.050
MRR 0.492 [0.472, 0.500]; flow 0/1/8/11/0; pipeline 0/0/12/7/1.				
L2 strength	20/20	0.500 [0.500, 0.500]	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/1/8/11/0; pipeline 0/0/0/0/0.				
lookalike.LX; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/–	–	0.000 [0.000, 0.000]	– / – / –
MRR 0.214 [0.188, 0.251]; flow 0/0/10/10/0; pipeline 0/0/0/0/0.				
L1 Q judge	14/–	–	0.881 [0.702, 1.000]	– / – / –
MRR 0.898 [0.745, 1.000]; flow 0/0/8/6/6; pipeline 0/0/0/0/0.				
L1 net	20/–	–	0.050 [0.000, 0.167]	0.000 / 0.050 / 0.950
MRR 0.368 [0.289, 0.461]; flow 0/0/10/10/0; pipeline 0/0/0/1/19.				
L1 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.100 / 0.900
MRR 0.283 [0.231, 0.339]; flow 0/0/10/10/0; pipeline 0/0/1/1/18.				
L1 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/10/10/0; pipeline 0/0/0/0/0.				
L2 corr.	20/–	–	0.050 [0.000, 0.167]	– / – / –
MRR 0.316 [0.240, 0.411]; flow 0/0/10/10/0; pipeline 0/0/0/0/0.				
L2 Q judge	9/–	–	0.907 [0.688, 1.000]	– / – / –
MRR 0.921 [0.732, 1.000]; flow 0/0/5/4/11; pipeline 0/0/0/0/0.				
L2 leak	20/–	–	0.150 [0.000, 0.318]	0.000 / 0.000 / 1.000
MRR 0.420 [0.306, 0.548]; flow 0/0/10/10/0; pipeline 0/0/0/0/20.				
L2 net	20/–	–	0.150 [0.000, 0.318]	0.000 / 0.000 / 1.000
MRR 0.398 [0.292, 0.524]; flow 0/0/10/10/0; pipeline 0/0/0/0/20.				
L2 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.600 / 0.400
MRR 0.237 [0.203, 0.279]; flow 0/0/10/10/0; pipeline 0/0/8/4/8.				
L2 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/10/10/0; pipeline 0/0/0/0/0.				
lookalike.LX-swap; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/–	–	0.000 [0.000, 0.000]	– / – / –
MRR 0.231 [0.194, 0.280]; flow 0/0/13/7/0; pipeline 0/0/0/0/0.				
L1 Q judge	7/–	–	0.881 [0.583, 1.000]	– / – / –
MRR 0.898 [0.643, 1.000]; flow 0/0/4/3/13; pipeline 0/0/0/0/0.				
L1 net	20/–	–	0.100 [0.000, 0.250]	0.000 / 0.000 / 1.000
MRR 0.399 [0.307, 0.509]; flow 0/0/13/7/0; pipeline 0/0/0/0/20.				
L1 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.000 / 1.000
MRR 0.277 [0.228, 0.336]; flow 0/0/13/7/0; pipeline 0/0/0/0/20.				
L1 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/13/7/0; pipeline 0/0/0/0/0.				
L2 corr.	20/–	–	0.100 [0.000, 0.250]	– / – / –
MRR 0.393 [0.296, 0.506]; flow 0/0/13/7/0; pipeline 0/0/0/0/0.				

Level / detector	Cov./n	AUROC [interval]	Top-1 [interval]	Unique / wrong / abstain
L2 Q judge	8/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/6/2/12; pipeline 0/0/0/0/0.				
L2 leak	20/–	–	0.100 [0.000, 0.250]	0.000 / 0.000 / 1.000
MRR 0.403 [0.316, 0.513]; flow 0/0/13/7/0; pipeline 0/0/0/0/20.				
L2 net	20/–	–	0.200 [0.048, 0.391]	0.000 / 0.050 / 0.950
MRR 0.438 [0.315, 0.582]; flow 0/0/13/7/0; pipeline 0/0/1/0/19.				
L2 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.650 / 0.350
MRR 0.237 [0.204, 0.276]; flow 0/0/13/7/0; pipeline 0/0/9/4/7.				
L2 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/13/7/0; pipeline 0/0/0/0/0.				
lookalike.LY; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/–	–	0.000 [0.000, 0.000]	– / – / –
MRR 0.188 [0.172, 0.210]; flow 0/0/15/5/0; pipeline 0/0/0/0/0.				
L1 Q judge	10/–	–	0.783 [0.528, 1.000]	– / – / –
MRR 0.867 [0.717, 1.000]; flow 0/0/8/2/10; pipeline 0/0/0/0/0.				
L1 net	20/–	–	0.100 [0.000, 0.250]	0.000 / 0.150 / 0.850
MRR 0.340 [0.241, 0.461]; flow 0/0/15/5/0; pipeline 0/0/2/1/17.				
L1 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.350 / 0.650
MRR 0.382 [0.326, 0.438]; flow 0/0/15/5/0; pipeline 0/0/3/4/13.				
L1 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/15/5/0; pipeline 0/0/0/0/0.				
L2 corr.	20/–	–	0.250 [0.059, 0.450]	– / – / –
MRR 0.520 [0.391, 0.655]; flow 0/0/15/5/0; pipeline 0/0/0/0/0.				
L2 Q judge	6/–	–	0.611 [0.233, 1.000]	– / – / –
MRR 0.742 [0.481, 1.000]; flow 0/0/3/3/14; pipeline 0/0/0/0/0.				
L2 leak	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.000 / 1.000
MRR 0.301 [0.254, 0.353]; flow 0/0/15/5/0; pipeline 0/0/0/0/20.				
L2 net	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.100 / 0.900
MRR 0.334 [0.280, 0.390]; flow 0/0/15/5/0; pipeline 0/0/1/1/18.				
L2 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.950 / 0.050
MRR 0.299 [0.251, 0.352]; flow 0/0/15/5/0; pipeline 0/0/8/11/1.				
L2 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/15/5/0; pipeline 0/0/0/0/0.				
lookalike.LY-swap; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/–	–	0.050 [0.000, 0.167]	– / – / –
MRR 0.276 [0.209, 0.371]; flow 0/0/17/3/0; pipeline 0/0/0/0/0.				
L1 Q judge	12/–	–	0.778 [0.533, 1.000]	– / – / –
MRR 0.814 [0.613, 1.000]; flow 0/0/11/1/8; pipeline 0/0/0/0/0.				
L1 net	20/–	–	0.050 [0.000, 0.167]	0.050 / 0.000 / 0.950
MRR 0.386 [0.312, 0.473]; flow 0/0/17/3/0; pipeline 0/0/1/0/19.				
L1 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.400 / 0.600
MRR 0.387 [0.330, 0.440]; flow 0/0/17/3/0; pipeline 0/0/5/3/12.				
L1 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/17/3/0; pipeline 0/0/0/0/0.				
L2 corr.	20/–	–	0.150 [0.000, 0.333]	– / – / –
MRR 0.420 [0.310, 0.548]; flow 0/0/17/3/0; pipeline 0/0/0/0/0.				
L2 Q judge	7/–	–	0.762 [0.444, 1.000]	– / – / –
MRR 0.796 [0.524, 1.000]; flow 0/0/6/1/13; pipeline 0/0/0/0/0.				
L2 leak	20/–	–	0.100 [0.000, 0.250]	0.000 / 0.050 / 0.950
MRR 0.367 [0.277, 0.478]; flow 0/0/17/3/0; pipeline 0/1/0/0/19.				
L2 net	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.150 / 0.850
MRR 0.313 [0.274, 0.356]; flow 0/0/17/3/0; pipeline 0/0/0/3/17.				
L2 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.900 / 0.100
MRR 0.355 [0.296, 0.412]; flow 0/0/17/3/0; pipeline 0/0/10/8/2.				
L2 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/17/3/0; pipeline 0/0/0/0/0.				
lookalike.LZ; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/–	–	0.050 [0.000, 0.167]	– / – / –
MRR 0.317 [0.235, 0.418]; flow 0/0/11/9/0; pipeline 0/0/0/0/0.				
L1 Q judge	12/–	–	0.944 [0.810, 1.000]	– / – / –
MRR 0.958 [0.857, 1.000]; flow 0/0/6/6/8; pipeline 0/0/0/0/0.				
L1 net	20/–	–	0.150 [0.000, 0.333]	0.000 / 0.050 / 0.950
MRR 0.386 [0.275, 0.517]; flow 0/0/11/9/0; pipeline 0/0/0/1/19.				
L1 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.000 / 1.000
MRR 0.249 [0.198, 0.308]; flow 0/0/11/9/0; pipeline 0/0/0/0/20.				

Level / detector	Cov./n	AUROC [interval]	Top-1 [interval]	Unique / wrong / abstain
L1 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/11/9/0; pipeline 0/0/0/0/0.				
L2 corr.	20/–	–	0.150 [0.000, 0.333]	– / – / –
MRR 0.376 [0.265, 0.511]; flow 0/0/11/9/0; pipeline 0/0/0/0/0.				
L2 Q judge	6/–	–	0.667 [0.250, 1.000]	– / – / –
MRR 0.806 [0.542, 1.000]; flow 0/0/2/4/14; pipeline 0/0/0/0/0.				
L2 leak	20/–	–	0.200 [0.048, 0.389]	0.000 / 0.000 / 1.000
MRR 0.429 [0.305, 0.572]; flow 0/0/11/9/0; pipeline 0/0/0/0/20.				
L2 net	20/–	–	0.100 [0.000, 0.250]	0.050 / 0.150 / 0.800
MRR 0.362 [0.268, 0.478]; flow 0/0/11/9/0; pipeline 0/0/0/4/16.				
L2 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.250 / 0.750
MRR 0.258 [0.212, 0.312]; flow 0/0/11/9/0; pipeline 0/0/1/4/15.				
L2 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/11/9/0; pipeline 0/0/0/0/0.				
lookalike.LZ-swap; declared 20, collected 20, missing 0, ineligible 0, eligible 20				
L1 corr.	20/–	–	0.050 [0.000, 0.167]	– / – / –
MRR 0.246 [0.189, 0.342]; flow 0/0/10/10/0; pipeline 0/0/0/0/0.				
L1 Q judge	8/–	–	0.583 [0.259, 0.889]	– / – / –
MRR 0.696 [0.455, 0.917]; flow 0/0/4/4/12; pipeline 0/0/0/0/0.				
L1 net	20/–	–	0.100 [0.000, 0.250]	0.000 / 0.000 / 1.000
MRR 0.353 [0.257, 0.473]; flow 0/0/10/10/0; pipeline 0/0/0/0/20.				
L1 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.000 / 1.000
MRR 0.211 [0.181, 0.251]; flow 0/0/10/10/0; pipeline 0/0/0/0/20.				
L1 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/10/10/0; pipeline 0/0/0/0/0.				
L2 corr.	20/–	–	0.150 [0.000, 0.333]	– / – / –
MRR 0.354 [0.243, 0.494]; flow 0/0/10/10/0; pipeline 0/0/0/0/0.				
L2 Q judge	5/–	–	0.800 [0.333, 1.000]	– / – / –
MRR 0.880 [0.600, 1.000]; flow 0/0/4/1/15; pipeline 0/0/0/0/0.				
L2 leak	20/–	–	0.250 [0.062, 0.450]	0.000 / 0.050 / 0.950
MRR 0.506 [0.381, 0.646]; flow 0/0/10/10/0; pipeline 0/1/0/0/19.				
L2 net	20/–	–	0.300 [0.111, 0.520]	0.050 / 0.050 / 0.900
MRR 0.512 [0.372, 0.666]; flow 0/0/10/10/0; pipeline 0/0/0/2/18.				
L2 paper	20/–	–	0.000 [0.000, 0.000]	0.000 / 0.150 / 0.850
MRR 0.254 [0.208, 0.309]; flow 0/0/10/10/0; pipeline 0/1/0/2/17.				
L2 strength	20/–	–	1.000 [1.000, 1.000]	– / – / –
MRR 1.000 [1.000, 1.000]; flow 0/0/10/10/0; pipeline 0/0/0/0/0.				

E.1 Prefix, missing-bid and breadth sensitivity

The following compact table provides the score trajectories. Full intervals, coverage, flow labels and reporting outcomes for every prefix/mask remain in `results/analysis/final_v1/all_cells.md`. A3 merges all registered breadth indices; it does not select the first subset after observing scores. The unavailable Gemma-E4B and Ministral-3B cells remain in the comprehensive table above. Mixed XX and XY treatments each use their own composition-matched honest control; their contrast does not isolate model identity.

Table 6: Prefix and missing-bid sensitivity for every available treatment comparison. Each entry is AUROC; exact intervals, exclusions and other metrics for every prefix/mask are in all_cells.md.

Cell / level / detector	Mask	n (full)	Prefix 50	100	200
core.P0/L1/net	complete	20	1.000	1.000	1.000
core.P0/L1/net	no_bid	20	1.000	1.000	1.000
core.P0/L1/correlation	complete	20	1.000	1.000	1.000
core.P0/L1/correlation	no_bid	20	1.000	1.000	1.000
core.P0/L2/net	complete	20	0.897	1.000	1.000
core.P0/L2/net	no_bid	20	0.890	1.000	1.000
core.P0/L2/correlation	complete	20	0.978	1.000	1.000
core.P0/L2/correlation	no_bid	20	0.978	1.000	1.000
core.P0/L2/leak	complete	20	0.953	1.000	1.000
core.P0/L2/leak	no_bid	20	0.953	1.000	1.000
core.C1-SS/L1/net	complete	20	0.882	0.915	0.912
core.C1-SS/L1/net	no_bid	20	0.882	0.915	0.917
core.C1-SS/L1/correlation	complete	20	0.968	0.978	0.988

Cell / level / detector	Mask	n (full)	Prefix 50	100	200
core.C1-SS/L1/correlation	no_bid	20	0.968	0.975	0.988
core.C1-SS/L2/net	complete	20	0.650	0.795	0.865
core.C1-SS/L2/net	no_bid	20	0.660	0.785	0.858
core.C1-SS/L2/correlation	complete	20	0.975	0.998	0.998
core.C1-SS/L2/correlation	no_bid	20	0.915	0.990	1.000
core.C1-SS/L2/leak	complete	20	0.522	0.562	0.640
core.C1-SS/L2/leak	no_bid	20	0.520	0.560	0.627
core.C1-SF/L1/net	complete	20	0.762	0.840	0.993
core.C1-SF/L1/net	no_bid	20	0.760	0.870	0.970
core.C1-SF/L1/correlation	complete	20	0.902	0.990	1.000
core.C1-SF/L1/correlation	no_bid	20	0.902	0.993	1.000
core.C1-SF/L2/net	complete	20	0.700	0.720	0.850
core.C1-SF/L2/net	no_bid	20	0.690	0.730	0.853
core.C1-SF/L2/correlation	complete	20	0.833	0.920	0.980
core.C1-SF/L2/correlation	no_bid	20	0.748	0.887	0.998
core.C1-SF/L2/leak	complete	20	0.650	0.642	0.645
core.C1-SF/L2/leak	no_bid	20	0.660	0.665	0.675
core.C1-FF/L1/net	complete	20	0.907	1.000	1.000
core.C1-FF/L1/net	no_bid	20	0.902	1.000	1.000
core.C1-FF/L1/correlation	complete	20	0.960	0.985	0.993
core.C1-FF/L1/correlation	no_bid	20	0.938	0.985	0.990
core.C1-FF/L2/net	complete	20	0.750	0.882	0.998
core.C1-FF/L2/net	no_bid	20	0.752	0.887	0.998
core.C1-FF/L2/correlation	complete	20	0.963	0.995	1.000
core.C1-FF/L2/correlation	no_bid	20	0.958	0.995	1.000
core.C1-FF/L2/leak	complete	20	0.580	0.740	0.868
core.C1-FF/L2/leak	no_bid	20	0.590	0.708	0.863
core.C1-NW/L1/net	complete	20	0.845	0.902	0.968
core.C1-NW/L1/net	no_bid	20	0.873	0.905	0.975
core.C1-NW/L1/correlation	complete	20	0.932	0.955	0.993
core.C1-NW/L1/correlation	no_bid	20	0.970	0.980	1.000
core.C1-NW/L2/net	complete	20	0.725	0.797	0.963
core.C1-NW/L2/net	no_bid	20	0.730	0.810	0.963
core.C1-NW/L2/correlation	complete	20	0.927	0.955	0.985
core.C1-NW/L2/correlation	no_bid	20	0.843	0.905	0.930
core.C1-NW/L2/leak	complete	20	0.598	0.613	0.682
core.C1-NW/L2/leak	no_bid	20	0.590	0.620	0.690
core.C2/L1/net	complete	20	0.662	0.688	0.610
core.C2/L1/net	no_bid	20	0.657	0.680	0.605
core.C2/L1/correlation	complete	20	0.333	0.407	0.463
core.C2/L1/correlation	no_bid	20	0.370	0.432	0.485
core.C2/L2/net	complete	20	0.445	0.470	0.655
core.C2/L2/net	no_bid	20	0.458	0.482	0.665
core.C2/L2/correlation	complete	20	0.410	0.502	0.495
core.C2/L2/correlation	no_bid	20	0.365	0.470	0.472
core.C2/L2/leak	complete	20	0.415	0.417	0.562
core.C2/L2/leak	no_bid	20	0.420	0.430	0.575
core.E1/L1/net	complete	20	0.578	0.585	0.675
core.E1/L1/net	no_bid	20	0.593	0.598	0.672
core.E1/L1/correlation	complete	20	0.450	0.500	0.482
core.E1/L1/correlation	no_bid	20	0.512	0.540	0.475
core.E1/L2/net	complete	20	0.458	0.665	0.700
core.E1/L2/net	no_bid	20	0.455	0.682	0.705
core.E1/L2/correlation	complete	20	0.440	0.500	0.415
core.E1/L2/correlation	no_bid	20	0.480	0.522	0.535
core.E1/L2/leak	complete	20	0.522	0.695	0.703
core.E1/L2/leak	no_bid	20	0.515	0.698	0.705
labs.gemma-12b.C1/L1/net	complete	14	0.765	1.000	1.000
labs.gemma-12b.C1/L1/net	no_bid	14	0.832	1.000	1.000
labs.gemma-12b.C1/L1/correlation	complete	14	0.995	1.000	1.000
labs.gemma-12b.C1/L1/correlation	no_bid	14	0.944	0.974	0.954
labs.gemma-12b.C1/L2/net	complete	14	0.699	0.883	0.995
labs.gemma-12b.C1/L2/net	no_bid	14	0.714	0.883	0.995
labs.gemma-12b.C1/L2/correlation	complete	14	0.842	1.000	1.000
labs.gemma-12b.C1/L2/correlation	no_bid	14	0.867	0.959	0.944
labs.gemma-12b.C1/L2/leak	complete	14	0.653	0.791	0.969

Cell / level / detector	Mask	n (full)	Prefix 50	100	200
labs.gemma-12b.C1/L2/leak	no_bid	14	0.781	0.796	0.949
labs.ministral-8b.C1/L1/net	complete	20	0.583	0.825	0.860
labs.ministral-8b.C1/L1/net	no_bid	20	0.590	0.820	0.855
labs.ministral-8b.C1/L1/correlation	complete	20	0.853	0.875	0.897
labs.ministral-8b.C1/L1/correlation	no_bid	20	0.853	0.875	0.897
labs.ministral-8b.C1/L2/net	complete	20	0.627	0.605	0.745
labs.ministral-8b.C1/L2/net	no_bid	20	0.618	0.613	0.743
labs.ministral-8b.C1/L2/correlation	complete	20	0.585	0.760	0.828
labs.ministral-8b.C1/L2/correlation	no_bid	20	0.585	0.760	0.828
labs.ministral-8b.C1/L2/leak	complete	20	0.590	0.637	0.675
labs.ministral-8b.C1/L2/leak	no_bid	20	0.588	0.650	0.675
sizes.qwen-4b.C1/L1/net	complete	20	0.858	0.955	0.993
sizes.qwen-4b.C1/L1/net	no_bid	20	0.858	0.955	0.993
sizes.qwen-4b.C1/L1/correlation	complete	20	0.760	0.915	0.978
sizes.qwen-4b.C1/L1/correlation	no_bid	20	0.760	0.915	0.978
sizes.qwen-4b.C1/L2/net	complete	20	0.718	0.865	0.943
sizes.qwen-4b.C1/L2/net	no_bid	20	0.723	0.860	0.945
sizes.qwen-4b.C1/L2/correlation	complete	20	0.723	0.838	0.955
sizes.qwen-4b.C1/L2/correlation	no_bid	20	0.723	0.838	0.955
sizes.qwen-4b.C1/L2/leak	complete	20	0.550	0.450	0.463
sizes.qwen-4b.C1/L2/leak	no_bid	20	0.552	0.448	0.470
mixed.C1-XX/L1/net	complete	20	0.780	0.912	0.965
mixed.C1-XX/L1/net	no_bid	20	0.802	0.912	0.958
mixed.C1-XX/L1/correlation	complete	20	0.943	0.985	0.990
mixed.C1-XX/L1/correlation	no_bid	20	0.948	0.988	0.990
mixed.C1-XX/L2/net	complete	20	0.573	0.657	0.885
mixed.C1-XX/L2/net	no_bid	20	0.580	0.675	0.900
mixed.C1-XX/L2/correlation	complete	20	0.875	0.920	0.998
mixed.C1-XX/L2/correlation	no_bid	20	0.885	0.917	0.998
mixed.C1-XX/L2/leak	complete	20	0.485	0.552	0.695
mixed.C1-XX/L2/leak	no_bid	20	0.490	0.545	0.705
mixed.C1-XY/L1/net	complete	20	0.950	0.963	1.000
mixed.C1-XY/L1/net	no_bid	20	0.960	0.970	1.000
mixed.C1-XY/L1/correlation	complete	20	0.980	1.000	1.000
mixed.C1-XY/L1/correlation	no_bid	20	0.980	1.000	1.000
mixed.C1-XY/L2/net	complete	20	0.818	0.948	1.000
mixed.C1-XY/L2/net	no_bid	20	0.830	0.950	1.000
mixed.C1-XY/L2/correlation	complete	20	0.970	1.000	1.000
mixed.C1-XY/L2/correlation	no_bid	20	0.968	1.000	1.000
mixed.C1-XY/L2/leak	complete	20	0.792	0.865	0.860
mixed.C1-XY/L2/leak	no_bid	20	0.762	0.865	0.868

F Statistics, provenance, and deviations

F.1 A1: warning removal

A1 compares matched N0/C1-SS/C1-NW triplets. Its profit contrast is the difference of additive N0-referenced joint-profit uplifts; the relative uplift companion is listed separately. All eligible triplets contribute to the non-judge estimates. Judge comparisons use only triplets judged successfully in all conditions. The L2 judge difference of 0.5 with interval $[0.5, 0.5]$ comes from a single triplet and is not precise evidence.

Table 7: A1: warning removed minus warning present. Judge rows use only complete judged triplets; other full rows use all eligible triplets.

Endpoint / subset	n	Difference [95% interval]
All: L1:net	20	0.0550 [-0.0425, 0.1725]
All: L2:net	20	0.0975 [-0.0051, 0.2300]
All: L2:leak	20	0.0425 [-0.1625, 0.2576]
All: L1:correlation	20	0.0050 [-0.0250, 0.0250]
All: L2:correlation	20	-0.0125 [-0.0675, 0.0100]
All: L1:Q judge	5	0.1600 [-0.4000, 0.6400]
All: L2:Q judge	1	0.5000 [0.5000, 0.5000]
All: joint_profit	20	-0.3478 [-0.4015, -0.2921]

Endpoint / subset	n	Difference [95% interval]
All: relative uplift	20	-1.0849 [-1.2533, -0.9299]
All: winning_price	20	-0.3335 [-0.3925, -0.2753]
All: pair_win_share	20	0.0706 [0.0540, 0.0879]
Judge L1 subset: L1:net	5	-0.0400 [-0.1600, 0.0000]
Judge L1 subset: L2:net	5	0.0400 [-0.1200, 0.4000]
Judge L1 subset: L2:leak	5	0.2800 [0.0000, 0.7200]
Judge L1 subset: L1:correlation	5	0.0000 [0.0000, 0.0000]
Judge L1 subset: L2:correlation	5	0.0000 [0.0000, 0.0000]
Judge L1 subset: L1:Q judge	5	0.1600 [-0.4000, 0.6400]
Judge L1 subset: joint_profit	5	-0.2905 [-0.3826, -0.1985]
Judge L1 subset: relative uplift	5	-0.9591 [-1.2861, -0.6699]
Judge L1 subset: winning_price	5	-0.2927 [-0.4227, -0.1860]
Judge L1 subset: pair_win_share	5	0.0607 [0.0267, 0.0972]
Judge L2 subset: L1:net	1	0.0000 [0.0000, 0.0000]
Judge L2 subset: L2:net	1	1.0000 [1.0000, 1.0000]
Judge L2 subset: L2:leak	1	0.0000 [0.0000, 0.0000]
Judge L2 subset: L1:correlation	1	0.0000 [0.0000, 0.0000]
Judge L2 subset: L2:correlation	1	0.0000 [0.0000, 0.0000]
Judge L2 subset: L2:Q judge	1	0.5000 [0.5000, 0.5000]
Judge L2 subset: joint_profit	1	-0.5287 [-0.5287, -0.5287]
Judge L2 subset: relative uplift	1	-1.0070 [-1.0070, -1.0070]
Judge L2 subset: winning_price	1	-0.5132 [-0.5132, -0.5132]
Judge L2 subset: pair_win_share	1	0.1158 [0.1158, 0.1158]

F.2 A2: honest placement swaps

Write s_{ab} for the score of strong firm a and fringe firm b. The registered symmetrised placement contrast is

$$T = \tfrac{1}{2}\{s_{00}(\text{cell}) - s_{00}(\text{twin}) + s_{01}(\text{twin}) - s_{01}(\text{cell})\}.$$

The descriptive within-original-cell interaction is $D = s_{00} - s_{01} - s_{10} + s_{11}$. The intervention changes fringe placement and therefore can change the rest of the market. It does not identify an isolated sameness mechanism. The two-sided test swaps cell/twin labels within each index and twin-pair block. Bootstrap sampling keeps blocks with the same market index together. The tables retain all registered endpoints, per-twin contrasts and corresponding successful-judge subsets. Candidate denominators are pair opportunities, not independent trials.

Table 8: A2: registered placement-swap tests. None rejects after Holm correction across the registered endpoints.

Endpoint	n	Mean T [95% interval]	Raw p	Holm p
L1:net	60	-0.00281 [-0.00872, 0.00254]	0.30401	1.00000
L2:net	60	0.00751 [-0.00636, 0.02201]	0.24581	1.00000
L2:leak	60	0.00616 [0.00003, 0.01211]	0.03771	0.26397
L1:correlation	60	-0.00034 [-0.01416, 0.01347]	0.97100	1.00000
L2:correlation	60	0.00494 [-0.00781, 0.01745]	0.51502	1.00000
L1:Q judge	15	0.04000 [-0.10833, 0.13553]	0.54763	1.00000
L2:Q judge	10	0.04500 [-0.01500, 0.13864]	0.38943	1.00000

Table 9: A2 descriptive companions: original-cell interaction D and per-twin-pair contrasts. Counts are contributing matched cell/twin blocks.

Endpoint / twin	n	Mean T [95% interval]	D [95% interval]
L1:net / pooled	60	-0.0028 [-0.0087, 0.0025]	-0.0091 [-0.0207, 0.0014]
L1:net / LX	20	-0.0085 [-0.0170, -0.0000]	-0.0117 [-0.0356, 0.0113]
L1:net / LY	20	-0.0007 [-0.0100, 0.0090]	-0.0095 [-0.0389, 0.0166]
L1:net / LZ	20	0.0008 [-0.0081, 0.0090]	-0.0061 [-0.0230, 0.0089]
L2:net / pooled	60	0.0075 [-0.0064, 0.0220]	0.0195 [-0.0090, 0.0505]
L2:net / LX	20	0.0093 [-0.0116, 0.0300]	0.0511 [0.0088, 0.0986]
L2:net / LY	20	0.0119 [-0.0109, 0.0335]	0.0100 [-0.0318, 0.0562]
L2:net / LZ	20	0.0014 [-0.0197, 0.0220]	-0.0026 [-0.0490, 0.0401]
L2:leak / pooled	60	0.0062 [0.0000, 0.0121]	0.0166 [0.0069, 0.0261]

Endpoint / twin	n	Mean T [95% interval]	D [95% interval]
L2:leak / LX	20	0.0042 [-0.0048, 0.0136]	0.0211 [-0.0003, 0.0411]
L2:leak / LY	20	0.0072 [-0.0036, 0.0179]	0.0151 [-0.0076, 0.0384]
L2:leak / LZ	20	0.0071 [-0.0017, 0.0163]	0.0136 [-0.0035, 0.0312]
L1:correlation / pooled	60	-0.0003 [-0.0142, 0.0135]	-0.0232 [-0.0484, 0.0021]
L1:correlation / LX	20	0.0489 [0.0308, 0.0661]	-0.0087 [-0.0373, 0.0182]
L1:correlation / LY	20	-0.0122 [-0.0462, 0.0279]	-0.0119 [-0.0727, 0.0538]
L1:correlation / LZ	20	-0.0377 [-0.0610, -0.0148]	-0.0489 [-0.1202, 0.0193]
L2:correlation / pooled	60	0.0049 [-0.0078, 0.0174]	-0.0160 [-0.0494, 0.0173]
L2:correlation / LX	20	0.0003 [-0.0158, 0.0153]	0.0113 [-0.0295, 0.0485]
L2:correlation / LY	20	0.0254 [-0.0008, 0.0523]	-0.0280 [-0.0749, 0.0193]
L2:correlation / LZ	20	-0.0109 [-0.0391, 0.0179]	-0.0313 [-0.0802, 0.0144]
L1:Q judge / pooled	15	0.0400 [-0.1083, 0.1355]	0.0080 [-0.0043, 0.0267]
L1:Q judge / LX	3	0.0000 [0.0000, 0.0000]	0.0167 [0.0000, 0.0500]
L1:Q judge / LY	6	0.2250 [0.0300, 0.4202]	-0.0050 [-0.0150, 0.0000]
L1:Q judge / LZ	6	-0.1250 [-0.3125, 0.0000]	0.0167 [0.0000, 0.0500]
L2:Q judge / pooled	10	0.0450 [-0.0150, 0.1386]	0.0560 [-0.0357, 0.2192]
L2:Q judge / LX	4	0.0037 [-0.0100, 0.0250]	-0.0225 [-0.0500, 0.0000]
L2:Q judge / LY	4	0.1313 [-0.0250, 0.4000]	0.1625 [-0.1000, 0.7000]
L2:Q judge / LZ	2	-0.0450 [-0.0500, -0.0400]	-0.0000 [-0.0000, 0.0000]

Table 10: A2 on each successful judge subset: descriptive detector companions, without an additional Holm family.

Judge / detector / twin	n	Mean T [95% interval]	D [95% interval]
L1/L1:net/pooled	15	-0.0004 [-0.0098, 0.0125]	-0.0064 [-0.0329, 0.0294]
L1/L1:net/LX	3	0.0029 [-0.0139, 0.0254]	0.0249 [-0.0063, 0.0663]
L1/L1:net/LY	6	-0.0065 [-0.0260, 0.0135]	-0.0458 [-0.1200, 0.0204]
L1/L1:net/LZ	6	0.0041 [-0.0054, 0.0161]	0.0172 [-0.0037, 0.0371]
L1/L2:net/pooled	15	0.0294 [0.0001, 0.0587]	0.0430 [-0.0255, 0.1052]
L1/L2:net/LX	3	0.0651 [-0.0198, 0.1198]	0.1393 [0.0410, 0.2532]
L1/L2:net/LY	6	0.0274 [-0.0213, 0.0765]	0.0364 [-0.0922, 0.1710]
L1/L2:net/LZ	6	0.0134 [-0.0011, 0.0277]	0.0013 [-0.0288, 0.0301]
L1/L2:leak/pooled	15	0.0138 [0.0007, 0.0263]	0.0222 [-0.0053, 0.0454]
L1/L2:leak/LX	3	0.0148 [-0.0127, 0.0395]	0.0504 [0.0337, 0.0780]
L1/L2:leak/LY	6	0.0153 [-0.0068, 0.0386]	0.0138 [-0.0468, 0.0679]
L1/L2:leak/LZ	6	0.0118 [-0.0072, 0.0269]	0.0164 [-0.0157, 0.0497]
L1/L1:correlation/pooled	15	-0.0078 [-0.0341, 0.0176]	-0.0210 [-0.0889, 0.0516]
L1/L1:correlation/LX	3	0.0556 [0.0127, 0.0952]	0.0412 [-0.0359, 0.0801]
L1/L1:correlation/LY	6	-0.0112 [-0.0519, 0.0310]	-0.0380 [-0.1649, 0.0533]
L1/L1:correlation/LZ	6	-0.0361 [-0.0684, -0.0116]	-0.0351 [-0.2002, 0.1231]
L1/L2:correlation/pooled	15	0.0142 [-0.0231, 0.0502]	-0.0287 [-0.0816, 0.0219]
L1/L2:correlation/LX	3	0.0085 [-0.0432, 0.0723]	-0.0000 [-0.0731, 0.0975]
L1/L2:correlation/LY	6	0.0442 [-0.0164, 0.1010]	-0.0361 [-0.1025, 0.0427]
L1/L2:correlation/LZ	6	-0.0130 [-0.0541, 0.0287]	-0.0357 [-0.1161, 0.0492]
L1/L1:judge:qwen-9b/pooled	15	0.0400 [-0.1083, 0.1355]	0.0080 [-0.0043, 0.0267]
L1/L1:judge:qwen-9b/LX	3	0.0000 [0.0000, 0.0000]	0.0167 [0.0000, 0.0500]
L1/L1:judge:qwen-9b/LY	6	0.2250 [0.0300, 0.4202]	-0.0050 [-0.0150, 0.0000]
L1/L1:judge:qwen-9b/LZ	6	-0.1250 [-0.3125, 0.0000]	0.0167 [0.0000, 0.0500]
L2/L1:net/pooled	10	-0.0025 [-0.0184, 0.0126]	-0.0468 [-0.0835, -0.0143]
L2/L1:net/LX	4	-0.0149 [-0.0420, 0.0148]	-0.0525 [-0.0934, 0.0158]
L2/L1:net/LY	4	0.0037 [-0.0403, 0.0259]	-0.0511 [-0.1669, 0.0294]
L2/L1:net/LZ	2	0.0100 [-0.0031, 0.0231]	-0.0268 [-0.0297, -0.0240]
L2/L2:net/pooled	10	0.0118 [-0.0256, 0.0428]	-0.0170 [-0.0831, 0.0517]
L2/L2:net/LX	4	-0.0027 [-0.1141, 0.0457]	0.0076 [-0.0933, 0.1339]
L2/L2:net/LY	4	0.0306 [-0.0001, 0.0852]	-0.0448 [-0.1308, 0.0678]
L2/L2:net/LZ	2	0.0031 [-0.0393, 0.0455]	-0.0105 [-0.1720, 0.1509]
L2/L2:leak/pooled	10	0.0005 [-0.0136, 0.0160]	-0.0141 [-0.0432, 0.0137]
L2/L2:leak/LX	4	-0.0022 [-0.0174, 0.0142]	-0.0046 [-0.0434, 0.0267]
L2/L2:leak/LY	4	0.0096 [-0.0235, 0.0460]	-0.0123 [-0.1029, 0.0635]
L2/L2:leak/LZ	2	-0.0123 [-0.0268, 0.0021]	-0.0366 [-0.0659, -0.0073]
L2/L1:correlation/pooled	10	0.0124 [-0.0326, 0.0543]	-0.0611 [-0.1287, -0.0047]
L2/L1:correlation/LX	4	0.0506 [0.0007, 0.0887]	-0.0092 [-0.0751, 0.0475]
L2/L1:correlation/LY	4	0.0008 [-0.0671, 0.0938]	-0.1173 [-0.3024, 0.0415]
L2/L1:correlation/LZ	2	-0.0411 [-0.1136, 0.0314]	-0.0524 [-0.0770, -0.0279]
L2/L2:correlation/pooled	10	0.0314 [-0.0159, 0.0776]	-0.0435 [-0.1275, 0.0403]

Judge / detector / twin	n	Mean T [95% interval]	D [95% interval]
L2/L2:correlation/LX	4	-0.0052 [-0.0768, 0.0345]	0.0007 [-0.2387, 0.1074]
L2/L2:correlation/LY	4	0.0276 [-0.0746, 0.1373]	-0.1512 [-0.2005, -0.1081]
L2/L2:correlation/LZ	2	0.1119 [0.0956, 0.1283]	0.0837 [0.0388, 0.1286]
L2/L2:judge:qwen-9b/pooled	10	0.0450 [-0.0150, 0.1386]	0.0560 [-0.0357, 0.2192]
L2/L2:judge:qwen-9b/LX	4	0.0037 [-0.0100, 0.0250]	-0.0225 [-0.0500, 0.0000]
L2/L2:judge:qwen-9b/LY	4	0.1313 [-0.0250, 0.4000]	0.1625 [-0.1000, 0.7000]
L2/L2:judge:qwen-9b/LZ	2	-0.0450 [-0.0500, -0.0400]	-0.0000 [-0.0000, 0.0000]

Endpoint	Same-model candidates	Different-model candidates
L1:net	0/120 (0.000%)	6/600 (1.000%)
L2:net	2/120 (1.667%)	10/600 (1.667%)
L2:leak	1/120 (0.833%)	1/600 (0.167%)

F.3 Judge failures and matched coverage

Qwen produced 278 valid and 322 invalid judgements out of 600; Gemma produced 5 valid and 35 invalid out of 40. Content failures were retained, not resubmitted for a preferred answer. Transport failures were reconciled and retried unchanged as recorded in the deviations. Breadth judging was cut for budget. Its absent rows are uncollected judge coverage, not judge scores of zero. The registered limit was 12,000 generated tokens per attempt with the registered retry; most failed calls ended with `length`, according to the raw-log observation in the final handover. This report does not infer a cause for every invalid output from the failure code alone.

The successful-output Qwen top-1 pattern in C1-SS was $0.808/0.881$ at L1/L2, versus SF $0.0476/0.0303$ and FF $0.100/0.0119$. The honest N0 strong reference pair received $0.952/1.000$. These are treatment-only available-output summaries, not the matched samples plotted in Figure 9. They are consistent with strong-pair preference in these cells, without establishing a universal rule or an internal reasoning mechanism. P0 is a counterexample to a universal strong-pair account: its matched judge AUROC is zero at each access level, with only 4 and 3 blocks. Tiny-sample percentile intervals can collapse and must not be read as certainty.

Table 11: Judge completion and content failures, including every attempted cell and access level. Invalid entries failed to return valid JSON under the registered budget and retry rule.

Judge / cell	Level	Total	Valid	Invalid JSON
gemma-12b/core.C1-SS	L1	10	3	7
gemma-12b/core.C1-SS	L2	10	1	9
gemma-12b/core.N0	L1	10	1	9
gemma-12b/core.N0	L2	10	0	10
qwen-9b/core.C1-FF	L1	20	10	10
qwen-9b/core.C1-FF	L2	20	14	6
qwen-9b/core.C1-NW	L1	20	12	8
qwen-9b/core.C1-NW	L2	20	10	10
qwen-9b/core.C1-SF	L1	20	14	6
qwen-9b/core.C1-SF	L2	20	11	9
qwen-9b/core.C1-SS	L1	20	13	7
qwen-9b/core.C1-SS	L2	20	7	13
qwen-9b/core.C2	L1	20	10	10
qwen-9b/core.C2	L2	20	9	11
qwen-9b/core.E1	L1	20	12	8
qwen-9b/core.E1	L2	20	7	13
qwen-9b/core.N0	L1	40	21	19
qwen-9b/core.N0	L2	40	15	25
qwen-9b/core.P0	L1	20	5	15
qwen-9b/core.P0	L2	20	4	16
qwen-9b/lookalike.LX	L1	20	14	6
qwen-9b/lookalike.LX	L2	20	9	11
qwen-9b/lookalike.LX-swap	L1	20	7	13
qwen-9b/lookalike.LX-swap	L2	20	8	12
qwen-9b/lookalike.LY	L1	20	10	10
qwen-9b/lookalike.LY	L2	20	6	14
qwen-9b/lookalike.LY-swap	L1	20	12	8
qwen-9b/lookalike.LY-swap	L2	20	7	13

Judge / cell	Level	Total	Valid	Invalid JSON
qwen-9b/lookalike.LZ	L1	20	12	8
qwen-9b/lookalike.LZ	L2	20	6	14
qwen-9b/lookalike.LZ-swap	L1	20	8	12
qwen-9b/lookalike.LZ-swap	L2	20	5	15

Table 12: Judge comparisons on identical successful matched blocks. Tiny or degenerate intervals do not imply precision.

Cell / level / detector	n	Detector AUROC [interval]	Judge AUROC [interval]
core.P0/L1/net	4	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
core.P0/L1/correlation	4	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
core.P0/L1/strength	4	0.500 [0.500, 0.500]	0.000 [0.000, 0.000]
core.P0/L2/net	3	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
core.P0/L2/correlation	3	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
core.P0/L2/strength	3	0.500 [0.500, 0.500]	0.000 [0.000, 0.000]
core.P0/L2/leak	3	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
core.C1-SS/L1/net	8	0.875 [0.600, 1.000]	0.320 [0.069, 0.617]
core.C1-SS/L1/correlation	8	1.000 [1.000, 1.000]	0.320 [0.069, 0.617]
core.C1-SS/L1/strength	8	0.500 [0.500, 0.500]	0.320 [0.069, 0.617]
core.C1-SS/L2/net	1	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
core.C1-SS/L2/correlation	1	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
core.C1-SS/L2/strength	1	0.500 [0.500, 0.500]	0.000 [0.000, 0.000]
core.C1-SS/L2/leak	1	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
core.C2/L1/net	4	0.250 [0.000, 1.000]	0.344 [0.000, 1.000]
core.C2/L1/correlation	4	0.500 [0.000, 1.000]	0.344 [0.000, 1.000]
core.C2/L1/strength	4	0.500 [0.500, 0.500]	0.344 [0.000, 1.000]
core.C2/L2/net	3	0.889 [0.750, 1.000]	0.333 [0.000, 1.000]
core.C2/L2/correlation	3	0.333 [0.000, 1.000]	0.333 [0.000, 1.000]
core.C2/L2/strength	3	0.500 [0.500, 0.500]	0.333 [0.000, 1.000]
core.C2/L2/leak	3	0.222 [0.000, 1.000]	0.333 [0.000, 1.000]
core.E1/L1/net	5	0.840 [0.438, 1.000]	0.600 [0.167, 1.000]
core.E1/L1/correlation	5	0.640 [0.188, 1.000]	0.600 [0.167, 1.000]
core.E1/L1/strength	5	0.500 [0.500, 0.500]	0.600 [0.167, 1.000]
core.E1/L2/net	1	1.000 [1.000, 1.000]	0.500 [0.500, 0.500]
core.E1/L2/correlation	1	0.000 [0.000, 0.000]	0.500 [0.500, 0.500]
core.E1/L2/strength	1	0.500 [0.500, 0.500]	0.500 [0.500, 0.500]
core.E1/L2/leak	1	1.000 [1.000, 1.000]	0.500 [0.500, 0.500]
core.C1-SF/L1/net	7	1.000 [1.000, 1.000]	0.602 [0.250, 0.880]
core.C1-SF/L1/correlation	7	1.000 [1.000, 1.000]	0.602 [0.250, 0.880]
core.C1-SF/L1/strength	7	0.500 [0.500, 0.500]	0.602 [0.250, 0.880]
core.C1-SF/L2/net	3	0.667 [0.000, 1.000]	0.667 [0.000, 1.000]
core.C1-SF/L2/correlation	3	0.889 [0.750, 1.000]	0.667 [0.000, 1.000]
core.C1-SF/L2/strength	3	0.500 [0.500, 0.500]	0.667 [0.000, 1.000]
core.C1-SF/L2/leak	3	0.111 [0.000, 1.000]	0.667 [0.000, 1.000]
core.C1-FF/L1/net	4	1.000 [1.000, 1.000]	0.438 [0.000, 1.000]
core.C1-FF/L1/correlation	4	0.938 [0.778, 1.000]	0.438 [0.000, 1.000]
core.C1-FF/L1/strength	4	0.500 [0.500, 0.500]	0.438 [0.000, 1.000]
core.C1-FF/L2/net	7	1.000 [1.000, 1.000]	0.551 [0.344, 0.812]
core.C1-FF/L2/correlation	7	1.000 [1.000, 1.000]	0.551 [0.344, 0.812]
core.C1-FF/L2/strength	7	0.500 [0.500, 0.500]	0.551 [0.344, 0.812]
core.C1-FF/L2/leak	7	0.939 [0.720, 1.000]	0.551 [0.344, 0.812]
core.C1-NW/L1/net	5	0.960 [0.840, 1.000]	0.480 [0.000, 1.000]
core.C1-NW/L1/correlation	5	1.000 [1.000, 1.000]	0.480 [0.000, 1.000]
core.C1-NW/L1/strength	5	0.500 [0.500, 0.500]	0.480 [0.000, 1.000]
core.C1-NW/L2/net	5	1.000 [1.000, 1.000]	0.800 [0.500, 1.000]
core.C1-NW/L2/correlation	5	1.000 [1.000, 1.000]	0.800 [0.500, 1.000]
core.C1-NW/L2/strength	5	0.500 [0.500, 0.500]	0.800 [0.500, 1.000]
core.C1-NW/L2/leak	5	0.640 [0.250, 1.000]	0.800 [0.500, 1.000]

Table 13: Judge Brier scores on matched discrimination samples; prevalence is the artificial treatment/control balance, not real-world collusion prevalence.

Cell / level / judge	n	Brier	Prevalence
core.P0/L1/qwen-9b	4	0.7700	0.500
core.P0/L2/qwen-9b	3	0.8503	0.500
core.C1-SS/L1/qwen-9b	8	0.5230	0.500
core.C1-SS/L2/qwen-9b	1	0.4625	0.500
core.C2/L1/qwen-9b	4	0.5472	0.500
core.C2/L2/qwen-9b	3	0.6054	0.500
core.E1/L1/qwen-9b	5	0.4267	0.500
core.E1/L2/qwen-9b	1	0.4100	0.500
core.C1-SF/L1/qwen-9b	7	0.3861	0.500
core.C1-SF/L2/qwen-9b	3	0.3630	0.500
core.C1-FF/L1/qwen-9b	4	0.4384	0.500
core.C1-FF/L2/qwen-9b	7	0.3758	0.500
core.C1-NW/L1/qwen-9b	5	0.4782	0.500
core.C1-NW/L2/qwen-9b	5	0.4284	0.500

F.4 Exploratory frozen-context probes

The complete-context slope was 0.0134737 $[-0.0035263, 0.0319480]$; 199 contexts were complete. One context was incomplete (market 7, round 130, seat 3). The leak uses 2997 observed bids, including observations from the incomplete context: raw 0.1481534 bits, corrected 0.0887914 bits, implemented $p = 0.001$. No resolved average linear response follows from the slope interval. A signed mean slope can cancel across contexts while mutual information detects distributional changes, but neither cancellation nor nonlinearity is demonstrated here.

The leak inference has a more fundamental limitation. Each cost-specific regenerated message is shared by five recipient bids. Individual-bid permutation within context breaks that bundle structure and can treat stochastic message variation as evidence against its null. The resulting null is anti-conservative for a claim of reproducible partner-cost transmission. The frozen estimate and implemented p-value are retained for transparency, with no causal claim. Repeated independently regenerated messages within cost/context and an inference procedure respecting that hierarchy are needed. Relabelling cost categories among whole bundles alone leaves context-conditional mutual information unchanged.

F.5 Pooled fingerprints: descriptive numerical caution

The frozen fingerprint analysis uses 1000 market-bootstrap draws, 99 permutations per draw and 999 for each point estimate. Some point estimates lie outside their percentile intervals. The report generator flags all such entries below; it does not alter or refit them. Bootstrap bias and Monte Carlo variation are possible explanations, not demonstrated causes. A numerical audit of their relative contributions remains open, so these intervals are not central evidence for mechanism classification.

Table 14: Pooled directional fingerprints in bits, retained as descriptive estimates. An asterisk flags a point outside its percentile interval; these intervals have not been re-estimated.

Cell / term	Markets	Estimate [95% interval]
core.N0/action	40	-0.00019 [-0.00015, 0.00346] *
core.N0/leak_to_member_1	40	-0.00010 [-0.00036, 0.00260]
core.N0/leak_to_member_2	40	0.00020 [-0.00005, 0.00384]
core.P0/action	20	0.30648 [0.29323, 0.31641]
core.P0/leak_to_member_1	20	0.30548 [0.28850, 0.32980]
core.P0/leak_to_member_2	20	0.33977 [0.32680, 0.36131]
core.C1-SS/action	20	0.09657 [0.08996, 0.12093]
core.C1-SS/leak_to_member_1	20	0.00855 [0.00725, 0.02324]
core.C1-SS/leak_to_member_2	20	0.00553 [0.00505, 0.01819]
core.C2/action	20	0.00128 [-0.00098, 0.00724]
core.C2/leak_to_member_1	20	-0.00067 [-0.00074, 0.00477]
core.C2/leak_to_member_2	20	0.00211 [0.00068, 0.01044]
core.E1/action	20	0.00160 [0.00086, 0.00769]
core.E1/leak_to_member_1	20	0.00471 [0.00233, 0.01251]
core.E1/leak_to_member_2	20	-0.00037 [-0.00082, 0.00750]

Cell / term	Markets	Estimate [95% interval]
core.C1-SF/action	20	0.14449 [0.14156, 0.20800]
core.C1-SF/leak_to_member_1	20	0.00821 [0.00907, 0.02701] *
core.C1-SF/leak_to_member_2	20	0.00591 [0.00677, 0.02706] *
core.C1-FF/action	20	0.35648 [0.34066, 0.45755]
core.C1-FF/leak_to_member_1	20	0.03533 [0.03043, 0.06108]
core.C1-FF/leak_to_member_2	20	0.03399 [0.03076, 0.05980]
labs.gemma-12b.N0/action	14	0.00123 [-0.00063, 0.00828]
labs.gemma-12b.N0/leak_to_member_1	14	-0.00109 [-0.00111, 0.00343]
labs.gemma-12b.N0/leak_to_member_2	14	-0.00057 [-0.00093, 0.00316]
labs.gemma-12b.C1/action	20	0.15676 [0.15692, 0.20001] *
labs.gemma-12b.C1/leak_to_member_1	20	0.02509 [0.02469, 0.05021]
labs.gemma-12b.C1/leak_to_member_2	20	0.02396 [0.02277, 0.04355]
labs.ministral-8b.N0/action	20	0.00464 [0.00276, 0.01030]
labs.ministral-8b.N0/leak_to_member_1	20	0.00083 [0.00003, 0.00545]
labs.ministral-8b.N0/leak_to_member_2	20	-0.00052 [-0.00102, 0.00436]
labs.ministral-8b.C1/action	20	0.05346 [0.04685, 0.07558]
labs.ministral-8b.C1/leak_to_member_1	20	0.00607 [0.00627, 0.01867] *
labs.ministral-8b.C1/leak_to_member_2	20	0.00833 [0.00780, 0.02171]
sizes.qwen-4b.N0/action	20	0.00245 [0.00162, 0.00922]
sizes.qwen-4b.N0/leak_to_member_1	20	-0.00038 [-0.00052, 0.00565]
sizes.qwen-4b.N0/leak_to_member_2	20	-0.00017 [-0.00069, 0.00628]
sizes.qwen-4b.C1/action	20	0.08314 [0.07147, 0.10169]
sizes.qwen-4b.C1/leak_to_member_1	20	0.00051 [0.00060, 0.01002] *
sizes.qwen-4b.C1/leak_to_member_2	20	0.00135 [0.00091, 0.01094]
sizes.ministral-3b.C1/action	2	0.06218 [0.02717, 0.12382]
sizes.ministral-3b.C1/leak_to_member_1	2	0.00330 [0.00192, 0.04274]
sizes.ministral-3b.C1/leak_to_member_2	2	-0.00611 [-0.00769, 0.03385]
mixed.D/action	20	-0.00175 [-0.00132, 0.00568] *
mixed.D/leak_to_member_1	20	0.00081 [-0.00023, 0.00701]
mixed.D/leak_to_member_2	20	0.00045 [-0.00029, 0.00993]
mixed.C1-XX/action	20	0.12049 [0.10443, 0.15526]
mixed.C1-XX/leak_to_member_1	20	0.01366 [0.01186, 0.02979]
mixed.C1-XX/leak_to_member_2	20	0.00496 [0.00454, 0.01660]
mixed.H-XYZZ/action	20	-0.00087 [-0.00118, 0.00305]
mixed.H-XYZZ/leak_to_member_1	20	-0.00002 [-0.00051, 0.00577]
mixed.H-XYZZ/leak_to_member_2	20	-0.00017 [-0.00041, 0.00313]
mixed.C1-XY/action	20	0.18722 [0.17532, 0.22114]
mixed.C1-XY/leak_to_member_1	20	0.00697 [0.00634, 0.01908]
mixed.C1-XY/leak_to_member_2	20	0.05460 [0.04467, 0.07799]
core.C1-NW/action	20	0.08148 [0.06583, 0.11067]
core.C1-NW/leak_to_member_1	20	0.00947 [0.00534, 0.02328]
core.C1-NW/leak_to_member_2	20	0.01562 [0.01350, 0.02799]
lookalike.LX/action	20	-0.00014 [-0.00056, 0.00222]
lookalike.LX/leak_to_member_1	20	0.00087 [0.00044, 0.00717]
lookalike.LX/leak_to_member_2	20	-0.00006 [-0.00044, 0.00396]
lookalike.LX-swap/action	20	-0.00021 [-0.00084, 0.00435]
lookalike.LX-swap/leak_to_member_1	20	-0.00180 [-0.00137, 0.00440] *
lookalike.LX-swap/leak_to_member_2	20	0.00100 [0.00042, 0.00491]
lookalike.LY/action	20	0.01095 [0.00339, 0.02283]
lookalike.LY/leak_to_member_1	20	0.00067 [-0.00012, 0.00521]
lookalike.LY/leak_to_member_2	20	0.00067 [-0.00014, 0.00675]
lookalike.LY-swap/action	20	0.00401 [0.00144, 0.00869]
lookalike.LY-swap/leak_to_member_1	20	0.00035 [-0.00011, 0.00445]
lookalike.LY-swap/leak_to_member_2	20	0.00128 [0.00027, 0.00550]
lookalike.LZ/action	20	-0.00149 [-0.00129, 0.00468] *
lookalike.LZ/leak_to_member_1	20	0.00039 [-0.00019, 0.00466]
lookalike.LZ/leak_to_member_2	20	-0.00134 [-0.00112, 0.00478] *
lookalike.LZ-swap/action	20	0.00048 [-0.00018, 0.00498]
lookalike.LZ-swap/leak_to_member_1	20	0.00000 [-0.00077, 0.00508]
lookalike.LZ-swap/leak_to_member_2	20	0.00074 [-0.00007, 0.00743]

F.6 Freeze, execution and reproducibility

The initial preregistration freeze was `56717aba03261552b9e09be4dded8b6e4cdc04b3`; the amendment freeze was `4c4b6f1911486d479bd043e3590e9e0f90f6a42d`. The final analysis executed at revision

7ca459c, with the documented junction-handling loader fix (deviation 5g). It did not run exactly the amendment checkout. The registered estimands are retained, and the executable change is disclosed separately. The manifest hash is a63ef1040a022c4d6a190fde1ae34dafdfefb13bd1626bdb57f19cdf2488a4ff.

The primary swap seed is 20261027; the main bootstrap seed is 20261028, the fingerprint seed is 20261029, and A1 uses 20261030. A2 uses 20261031 for swaps and 20261032 for bootstrap.

Input SHA-256 digests. These are file-byte digests; the manifest's canonical content hash above is a different convention. results/analysis/final_v1/results.json
36804b9c365f98bb03bc92d7ab44b403a1ed6a6cf274339a968e035df5fdc672
results/analysis/final_v1/all_cells.md
fc7213378b13c7b9dcd57bed5305782d7ed56d03fbbc145b76d0bfe1d2cb7c08
experiments/manifests/collection_v2.json
34799532e08c197d1797476de2624943b97c919b4d849d7af52703275553f484
results/analysis_runs.json
e298ce13a69ce4ff1f814986ef012ce5e13631db1e8a8d80d2ef0adfc910dd4d
results/llm/judge_index_registered.json
cc0050be561b8df157fd97db239f44be06404dbe0d2917ac9f6fe19aa8e50069
results/llm/probe_index.json
af80667eda2a49569ee9c39fc7503e18840f0088d6bce3282540fa0e7a953175
src/collusion/llm/prompts.py
93e585cd275f28e827e1395bfacd47cde439f6148e4fccaa12ba4a685e5ab11a
src/collusion/analysis/io.py
3fb9b6561ab8d18f1e6bf1ba5b5a29c5b47ad3e3d5c0e08f1db5535031bdc5e5
results/analysis/judge24k_v23/results.json
5eebad9350886fb36ca85107f5f02d351006bda69c98353a8fd1dea62a947b48

The digests are of the current files. src/collusion/analysis/io.py differs from the version that executed final_v1 (revision 7ca459c, digest 4df9668f...7455 with Windows line endings) only by the optional –run–index argument added for the addenda (commit 63cb9f4), which does not change any statistic. The last digest is the judge-budget cohort used in Figure 9.

The following command reproduces the final analysis from released local inputs. It is documented here, not rerun during report preparation.

```
.venv/Scripts/python experiments/analyze_llm.py --unblind
  --input results --manifest experiments/manifests/collection_v2.json
  --analysis-freeze-commit 4c4b6f1911486d479bd043e3590e9e0f90f6a42d
  --judges results/llm/judge_index_registered.json
  --probes results/llm/probe_index.json --name final_v1 --workers 4
```

Join the displayed lines into one command. From the repository root, the report-only command is:

```
.venv/Scripts/python experiments/report_figures.py
```

The optional –quick flag produces identical report numbers: the script reads frozen estimates and performs no resampling. It writes PDF/PNG figures and a numerical CSV/JSON to results/report_figures/, copies PDFs to report/figures/, and generates the numerical LaTeX tables. results/report_figures/provenance.json records the source digest, analysis settings and bootstrap-weight digest. In report/, compile with automatic package installation disabled:

```
pdflatex --disable-installer -interaction=nonstopmode main.tex
bibtex --disable-installer main
pdflatex --disable-installer -interaction=nonstopmode main.tex
pdflatex --disable-installer -interaction=nonstopmode main.tex
```

F.7 Addenda v2.2–v2.3: replication and judge budget

Both addenda were committed before their data (63cb9f4, 9d759d0). The replication cohort re-declares the registered cells core.E1, core.C2 and core.C1-SS on markets 20–39 with experiments/cohort_index.

Table 15: Out-of-sample replication (Addendum v2.2). Original: registered markets 0–19; replication: fresh treatment markets 20–39 against the registered N0 markets 20–39. Estimates with 95% block-bootstrap intervals (20 matched blocks). Profit is the paired joint-profit difference versus N0.

Cell	Sample	L2 net AUROC	L2 net − corr.	Joint profit vs N0
E1	original	0.700 [0.529, 0.877]	0.285 [0.062, 0.515]	−0.116 [−0.137, −0.095]
E1	replication	0.680 [0.505, 0.830]	0.265 [0.070, 0.445]	−0.084 [−0.109, −0.060]
C2	original	0.655 [0.467, 0.833]	0.160 [−0.125, 0.453]	0.399 [0.364, 0.434]
C2	replication	0.752 [0.582, 0.905]	0.122 [−0.118, 0.358]	0.283 [0.179, 0.371]
C1-SS	original	0.865 [0.734, 0.968]	−0.133 [−0.265, −0.031]	0.144 [0.094, 0.192]
C1-SS	replication	0.907 [0.802, 0.985]	−0.088 [−0.190, −0.015]	0.173 [0.123, 0.221]

`py`, which copies each registered manifest entry and changes only cohort, indices, control and reason. Its matched control is the registered N0 run on markets 20–39. Treatment runs: `results/llm/rep_{E1,C2,C1-SS}_v22_deviation_replication`, each complete with 4,000 rounds. Pre-specified verdicts: R1 (E1, L2 net minus correlation above zero) replicated; R1b (the same at L1) did not; R2 (C2 joint-profit uplift above zero) replicated; R4 (C1-SS L2 net AUROC above 0.5) replicated (0.9075 $[0.8025, 0.985]$). The judge check used the registered prompt, seed and sampling with a 24,000-token limit (recorded in each run's configuration and reproducibility record). Eight second-attempt streams broke server-side after 400–785 s without a reply; they were released with an audit record and resubmitted unchanged, as for the registered judges. Of 60 judgements, 54 returned valid output and 6 invalid JSON; the registered 12,000-token judge had 27 valid outputs on the same 60 markets. A check cohort on the registered markets 0–19 reproduced `final_v1` exactly for correlation, strength and paper scores and within permutation noise for net and leak (largest AUROC difference 0.0075), because permutation seeds depend on the cohort name. Commands (one line each):

```
.venv/Scripts/python experiments/analyze_llm.py --unblind --input results
  --run-index results/analysis_runs_v22.json
  --manifest experiments/manifests/collection_v2.json
  --analysis-freeze-commit 4c4b6f1911486d479bd043e3590e9e0f90f6a42d
  --name replication_v22 --workers 3
.venv/Scripts/python experiments/analyze_llm.py --unblind --input results
  --run-index results/analysis_runs_v23.json
  --manifest experiments/manifests/collection_v2.json
  --judges results/llm/judge_index_24k.json
  --analysis-freeze-commit 4c4b6f1911486d479bd043e3590e9e0f90f6a42d
  --name judge24k_v23 --workers 3
.venv/Scripts/python experiments/addendum_summary.py
  --replication results/analysis/replication_v22/results.json
  --judge results/analysis/judge24k_v23/results.json
  --out results/analysis/addendum_summary
```

F.8 Deviations and collection order

The following summary follows the appended record in `docs/PREREG.md`; per-run metadata and `results/llm/reconciliation_log.json` retain affected attempts.

1. Warm-up transport handling was corrected before any bid was collected. The stopped pilot attempt was retained.

2. A versioned parser accepted a single Markdown code block around valid JSON for Gemma/Ministral. Initial invalid A2 attempts were retained and recollected as labelled cohorts. Core Qwen and A1 kept the original parser.

3. Concurrent-request caching, lost judge replies, streaming, timeouts and Windows checkpoint replacement were repaired. Unread transport replies were reconciled and requests resumed unchanged; answers were not selected on content.

Table 16: Addenda: AUROC versus matched N0 for every detector, original (markets 0–19) and replication (markets 20–39), with eligible markets n; 95% intervals are in `addendum_summary/summary.json`. Bottom: judge token-budget check.

Cell	Sample	L1 net	L1 corr.	L1 paper	L2 net	L2 corr.	L2 paper	L2 leak	n
E1	original	0.675	0.482	0.620	0.700	0.415	0.652	0.703	20
E1	replication	0.525	0.430	0.492	0.680	0.415	0.620	0.745	20
C2	original	0.610	0.463	0.710	0.655	0.495	0.887	0.562	20
C2	replication	0.598	0.642	0.585	0.752	0.630	0.760	0.573	20
C1-SS	original	0.912	0.988	0.885	0.865	0.998	0.882	0.640	20
C1-SS	replication	0.945	0.993	0.932	0.907	0.995	0.887	0.618	20

Judge (L2, markets 0–19)	valid	C1-SS AUROC	top-1 C1-SS	top-1 C1-SF
12,000 tokens (registered)	27/60	0.000 (1 block)	0.881	0.030
24,000 tokens (Addendum v2.3)	54/60	0.625 (17 blocks)	0.921	0.020

4. Disk exhaustion interrupted collection. Raw data moved to another drive through a junction, storage guards were added, and orphaned requests were reconciled. New run identifiers reused completed cached requests.

5. Later recovery addressed drive-specific guards, preemption, warm-up caps, ledger contention, call ceilings and judge timeouts. The loader's containment check was changed to accept the raw-data junction. The original budget cap was exceeded and raised; Modal billing was USD 70.35 before the addenda and USD 76.30 after them (deviation 7).

6. Breadth judges were cancelled for budget, while core, A1, A2 and the Gemma counter-judge were completed as attempted judgements. Memory failures and interrupted judge streams were reconciled; autoscaling was pinned for the batch. Content-invalid outputs remained failures. The counter-judge used its registered market subset and the fenced parser.

7. For the addenda, eight broken judge streams were released and resubmitted unchanged, and the budget stop was made tiered (judge before C1-SS); no tier was reached. The cohort index was regenerated as full copies of registered entries before any replication statistic.

The amendment was recorded before LLM outcome analysis. Actual A1 collection started alongside the core; A2 and breadth work were interleaved after warm-up, rather than following the proposed fully sequential order. The report retains this operational ordering and the parser cohorts instead of describing an uninterrupted frozen-code collection.

G Limitations and Dual-Use / Ethical Considerations

Instructed coordination and treatment attribution. The cartel is assigned by prompt; this is not evidence of emergent collusion among independently deployed agents. The treatment combines a joint-profit objective, partner naming, secrecy, a surveillance warning, and, in the channel condition, private communication. Their effects cannot generally be separated. The warning-removal and no-channel comparisons vary specific components, without identifying all interactions in the bundle. Poor compliance and refusal remain part of the assigned treatment outcomes.

Synthetic environment and external validity. No real procurement participants are involved. The firms, private costs, contracts, and payoffs belong to a synthetic game. Its discrete bids, independently redrawn costs, asymmetric firms, and fixed horizon simplify real procurement. Real bidders can have correlated costs, durable capacity constraints, changing strategies, and legitimate information exchange. Model family and size comparisons test robustness in this environment, not intrinsic propensities to collude. Mixed-population contrasts do not isolate identity from market strength; A2 identifies a specified placement change, including its effects on the rest of the market.

Dependence, intent, and harm. An influence score measures statistical dependence, not agreement, intent, illegality, or harm. A leak can reflect unilateral espionage, legitimate disclosure, or a cooperative cartel. A one-way pattern does

not distinguish all such cases, and nondetection does not establish absence. Realised profit uplift and winning-price differences are manipulation checks in the game; they do not prove a real-world welfare loss. Audits must retain unclear cases and abstentions instead of forcing a substantive accusation.

Null validity and finite data. Within-state permutation tests assume exchangeability that can fail when bidders adapt to omitted history, drift, or their own cumulative outcomes. The correction reduces estimated null bias under the specified shuffle; it does not provide a universal false-alarm guarantee. Sparse conditional tables, limited independent markets, missing bids, and model-specific parse failures affect precision and generalisation. The matched primary test has its own condition-label exchangeability null. Secondary intervals are pointwise, and error control for one detector or test family does not extend to all reported comparisons. Pooled fingerprints and repeated probe draws must not be counted as additional independent markets.

False accusations and human review. A false alarm can wrongly implicate an honest pair responding to shared history or similar incentives. These scores should support investigation with contextual evidence and human review, not automated sanctions. The evaluation must present honest controls, wrong-pair reports, abstention, and uncertainty alongside successful rankings. A judge's answer is another fallible baseline; family overlap and incomplete coverage limit its role as independent corroboration.

Private-input access and privacy. L2 assumes an authorised auditor can inspect private costs, an informational advantage that need not exist for a competition authority. A platform might hold agent inputs without being entitled to reuse or disclose them for every purpose. Deployment would require a legitimate access basis, purpose limitations, access controls, and protection of commercially sensitive data. The synthetic inputs here do not resolve those governance questions.

Dual use and adaptive concealment. Publishing detector strengths and blind spots could help design cartels or evade detection. In particular, behaviour-only auditing can miss dependence absorbed by the conditioning state, and an adaptive agent may deliberately suppress detectable private-cost dependence. This experiment does not test steganographic channels or establish robustness against agents optimising against the audit. The defensive value is to expose such limits and support better evaluation, rather than to offer a covert-coordination recipe.

Compute and reproducibility. Collection used cloud-hosted open-weight models; interruptions, retries, idle time, and judging contribute to resource use. The registered budget cap was exceeded, so the initial budget is not an account of actual spend. Total Modal billing for the project was USD 70.35 before the addenda (deployment, pilot, collection, judging and probes) and USD 76.30 metered after them, of which USD 76.03 came from credits (24 October 2026, all runs ended). Inference seeds and pinned configurations do not ensure identical outputs on re-collection. Logs and released data support analysis reproduction; unreconciled attempts, parser changes, and missing judgements must remain visible in that record.

H LLM usage statement

Claude Code was used for planning, reviews, commits, the addenda code and runs, and the final revision of this report. Codex models wrote code in implementation rounds, reviewed the frozen results, and prepared the first report draft. Marvin Kemper made the research decisions and reviewed the work. The LLM bidders and judges described in the methods are experimental subjects and baselines, distinct from the tools used to develop the study. AI assistance does not establish the correctness of a citation, numerical result, or interpretation; the final claims remain Marvin's responsibility. **[TODO: Marvin to complete the final substantive review and revise the prose in his own words before submission, as encouraged by the report template.]**